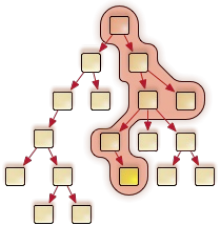
# Understanding Software Merge through Empirical Studies and EEG Data

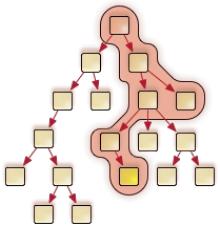
**Prof. Dr. Kleinner Silva Farias de Oliveira**
Postgraduate Program in Applied Computing – PPGCA
University of Vale do Rio dos Sinos
https://kleinnerfarias.github.io/
kleinnerfarias@unisinos.br





# Understanding Software Merge through Empirical Studies and EEG Data

**Prof. Dr. Kleinner Silva Farias de Oliveira**
Postgraduate Program in Applied Computing – PPGCA
University of Vale do Rio dos Sinos
https://kleinnerfarias.github.io/
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# In this talk…

- Why should we investigate?
- How do we investigate?
- What are the main results?

# Why…

Software merge…

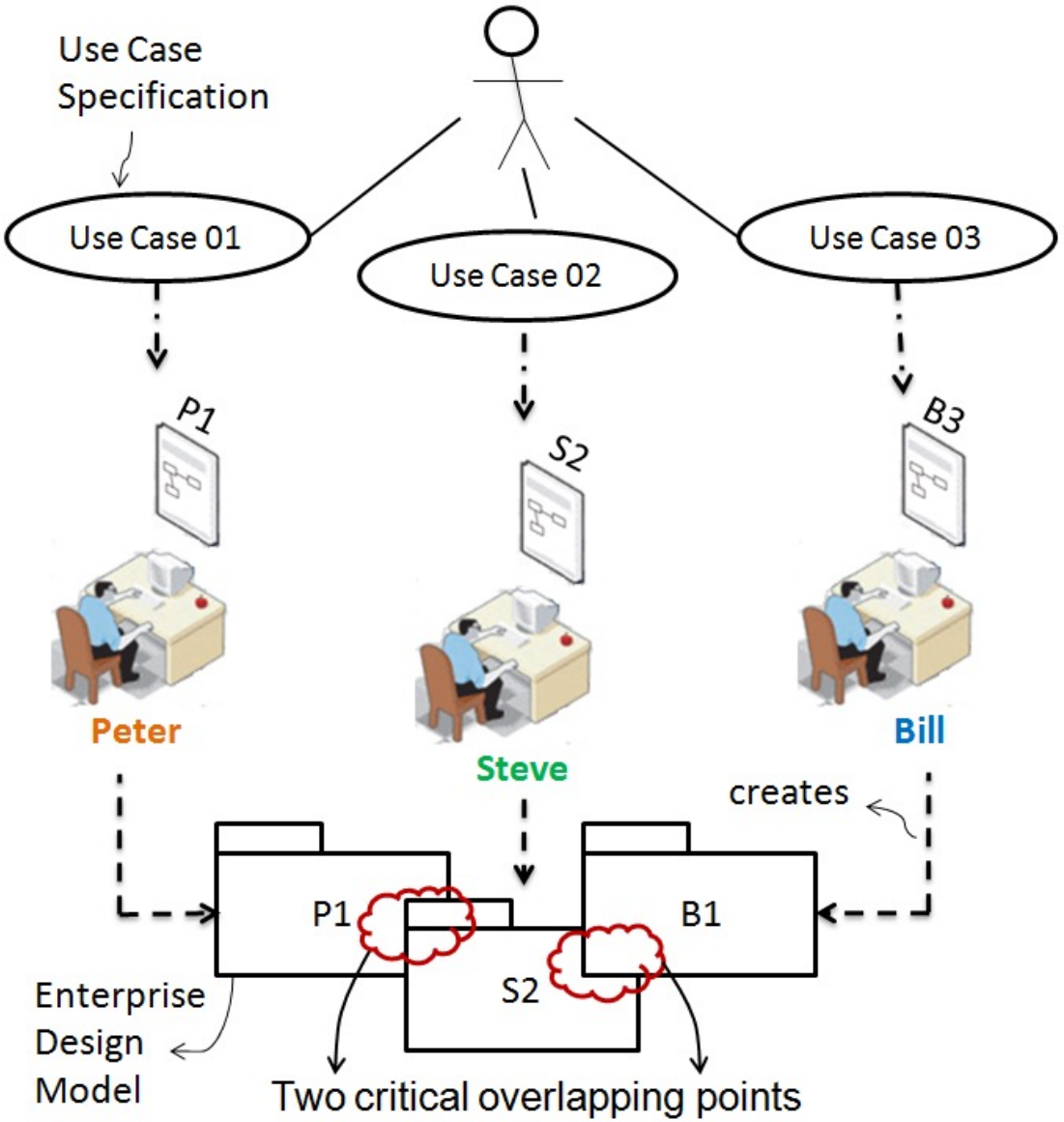
…really hard

…error-prone

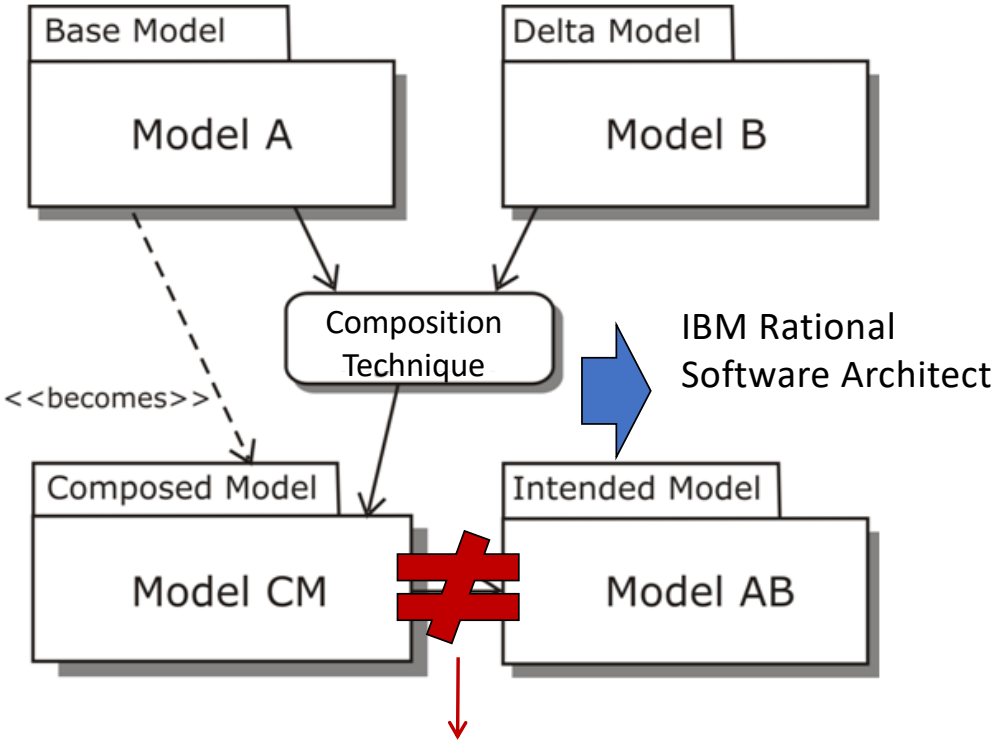
…time-consuming

…the current literature fails to provide empirical evidence

# Motivation


Use Case Specification
Use Case 01
Use Case 02
Use Case 03
P1
S2
B3
Peter
Steve
Bill
creates
P1
B1
S2
Enterprise Design Model
Two critical overlapping points

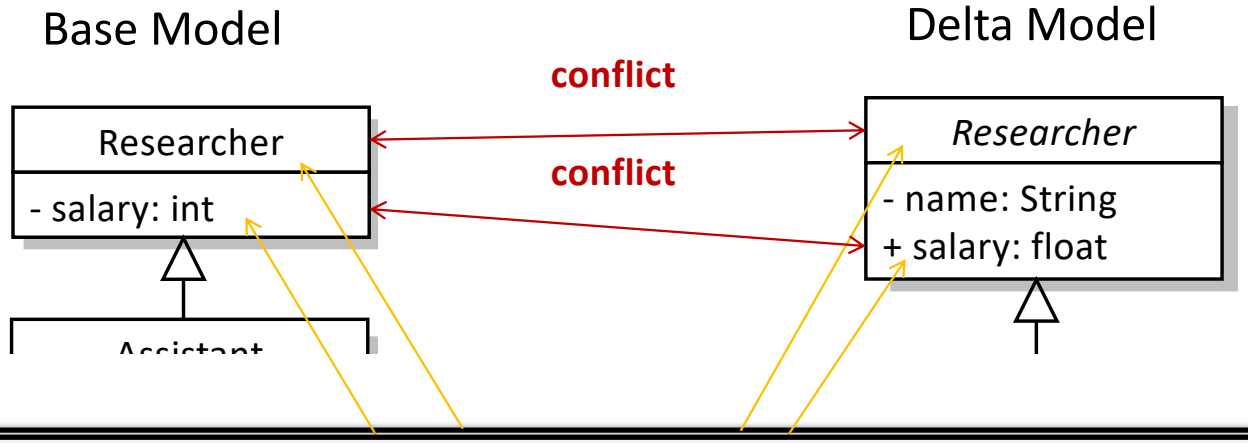
# Introduction






There is a difference because
the input models are in conflict in some way.

# Example: applying composition techniques




Attributes with equal names, but different types: INT vs FLOAT and

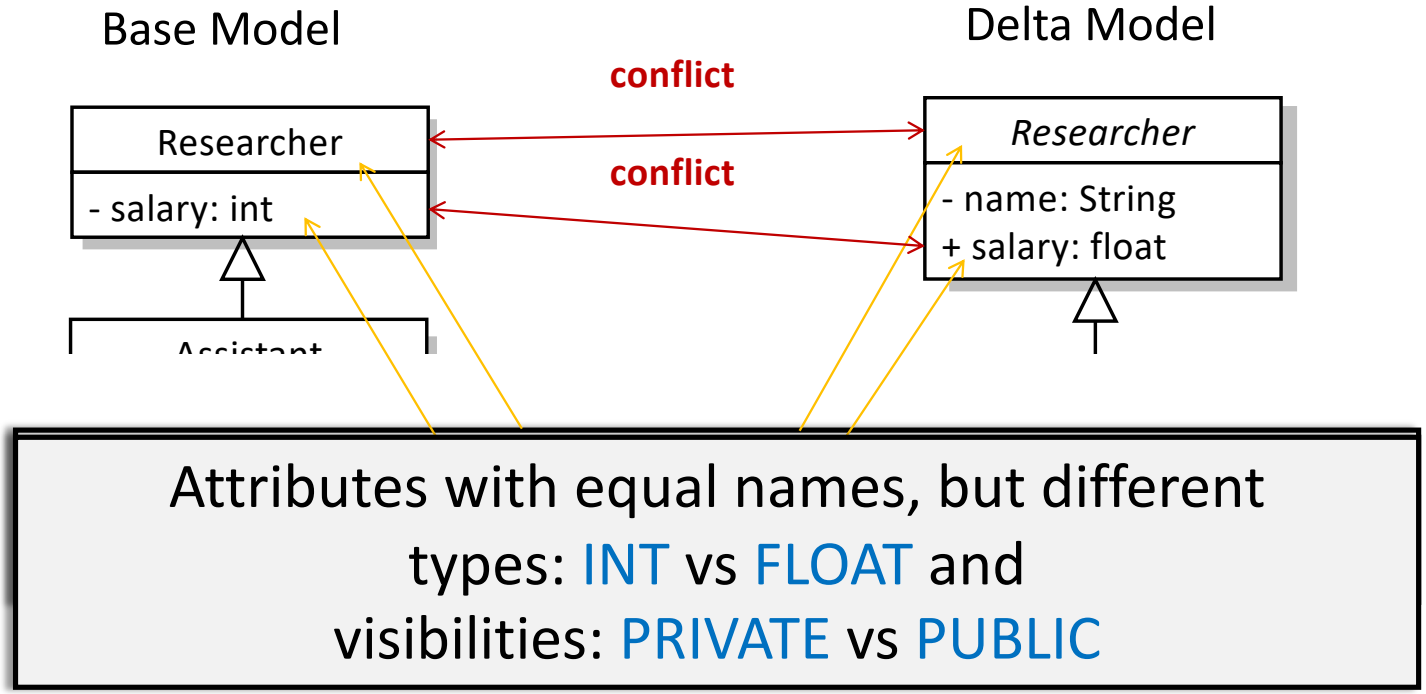


Developer needs to compose two input models.

*e input models*
*flict in some way.*


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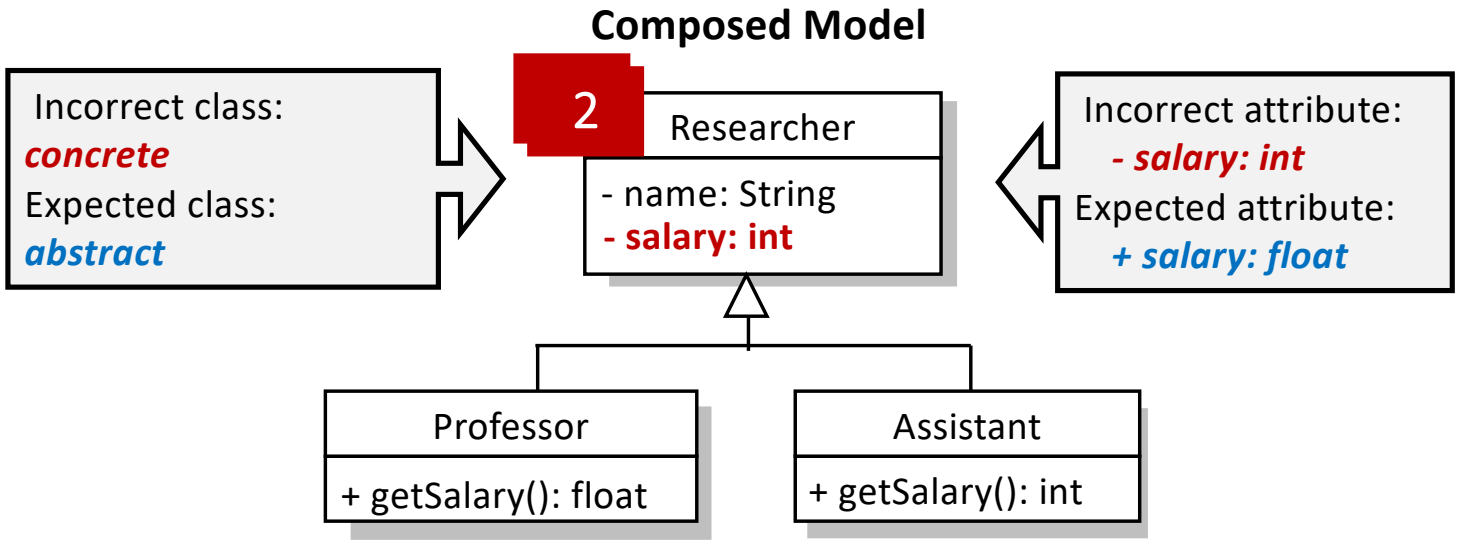
# Example: applying composition techniques




In practice, these conflicts are not always solved. As a result, instead of producing the output intended model, as would be expected, the techniques end up producing an output composed model with inconsistencies. To produce the intended model, the developer should necessarily invest some extra to detect and resolve all inconsistencies.


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# Example: detecting inconsistencies



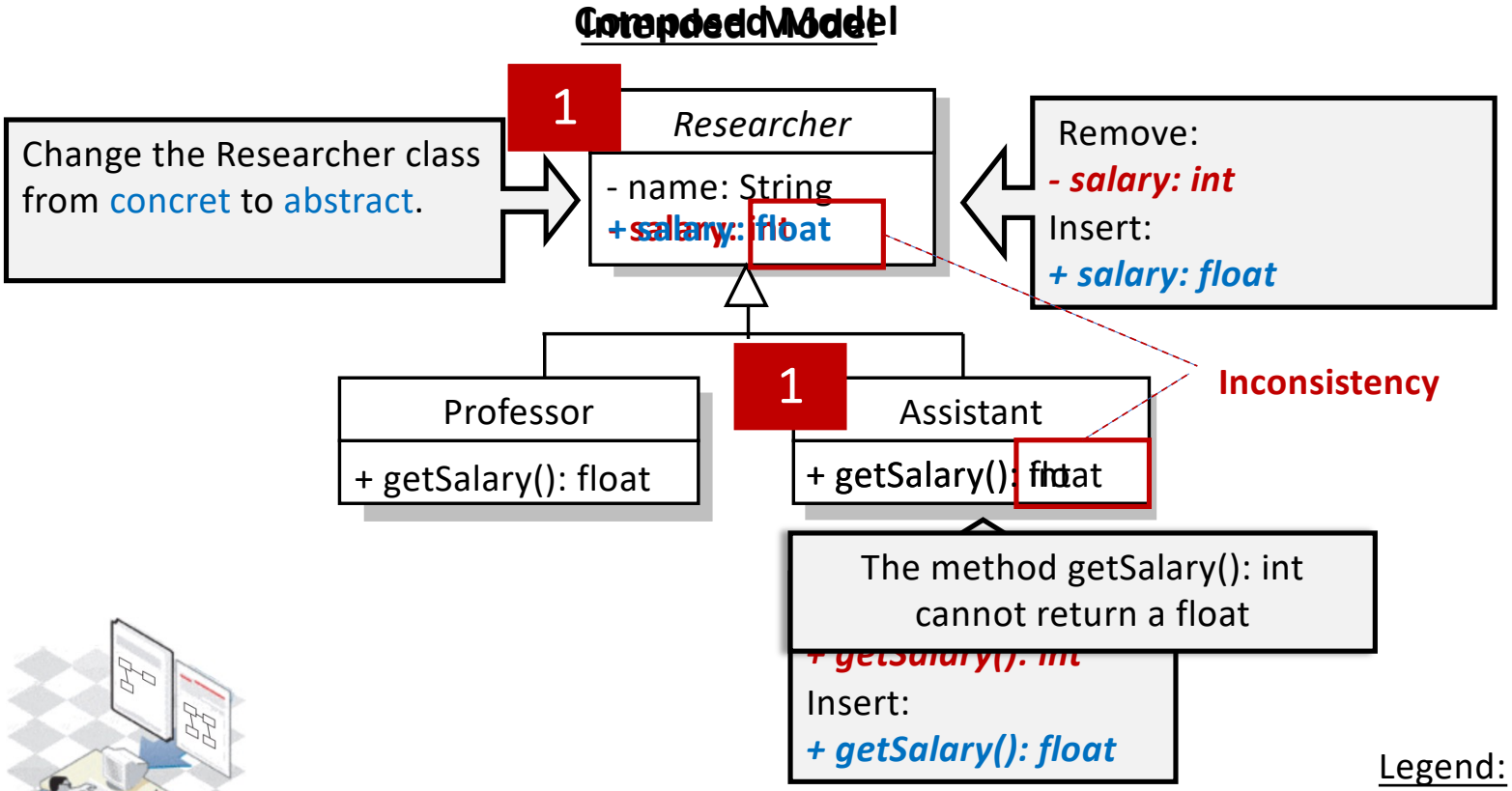



Legend:

Unexpected

Expected

# Example: resolving inconsistencies






# How: Case Study in Industry

…quantifying 297 compositions

…manipulating UML design models with more than 2500 classes in total

…observing software developers for 56 weeks

…working with real-world design models

…exploring technical and human factor in industry

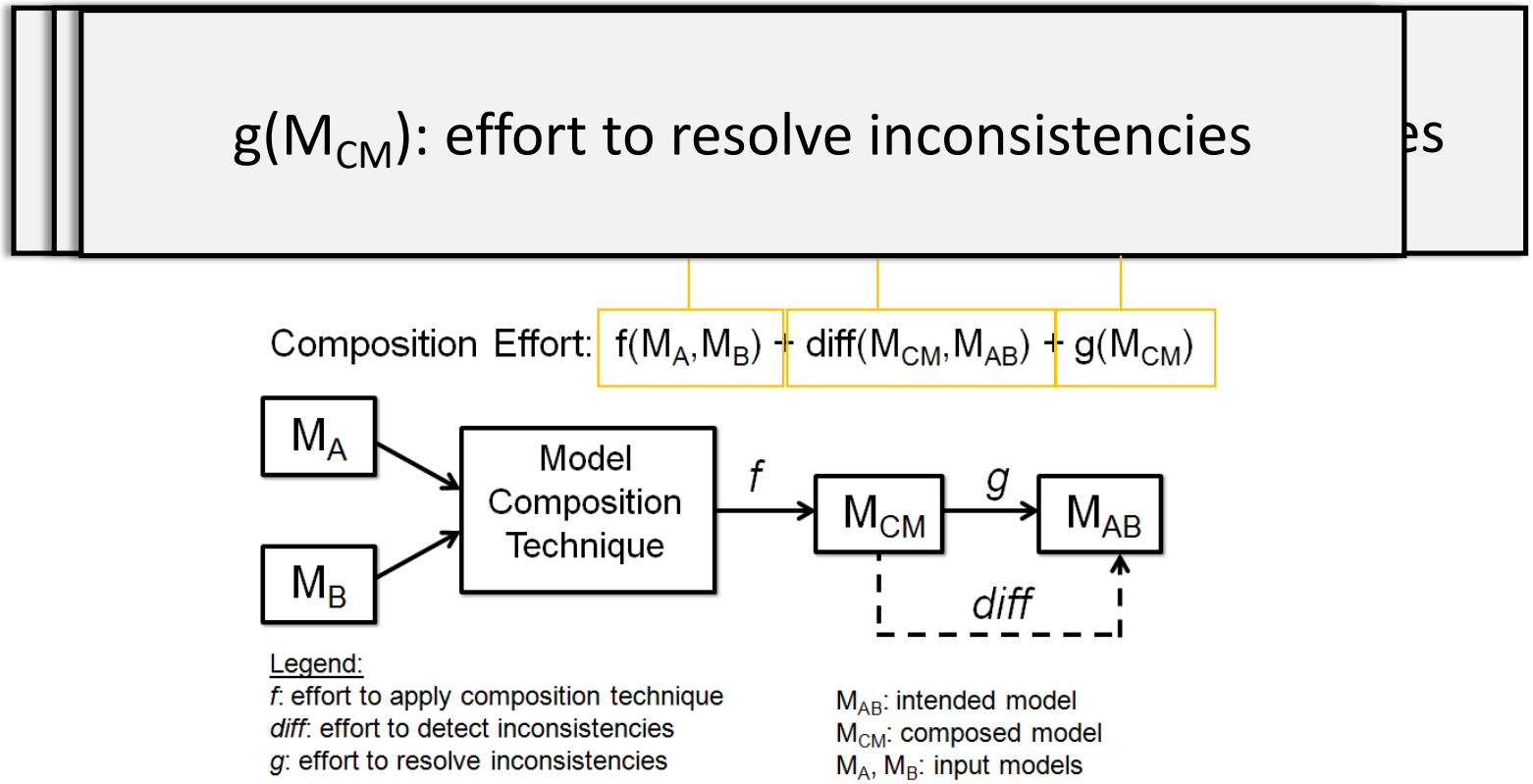
…case study performed in the field of Oil and Gas industry

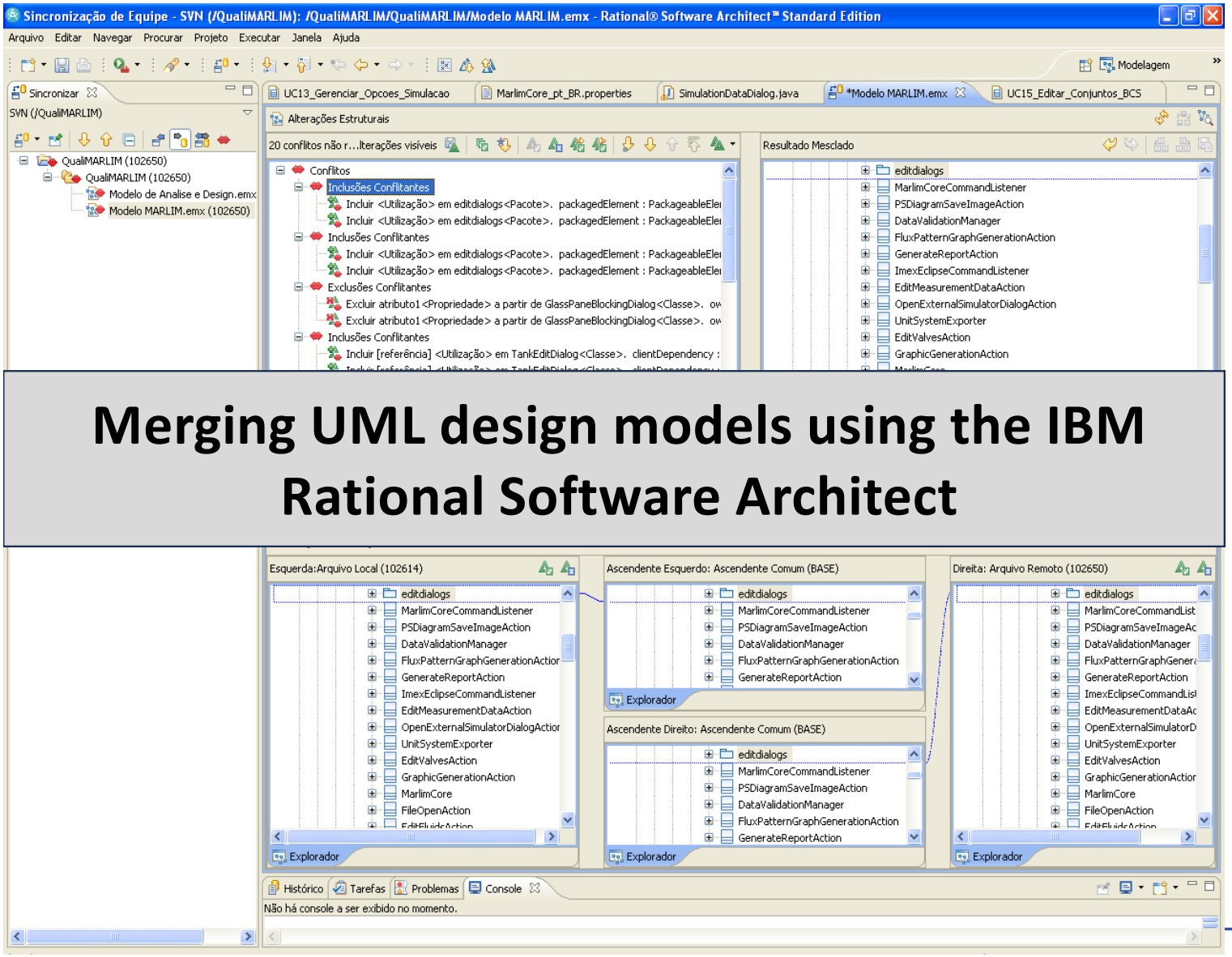
# Study Methodology

- **IBM Rational Software Architect**
  1. A robust and semi-automated model composition tool
  2. The tight integration with the Eclipse platform
  3. It was being used and adopted in previous successful projects
- **Model Composition Effort =**
  - effort to apply the model composition technique +
  - effort to detect the inconsistencies +
  - effort to resolve the inconsistencies

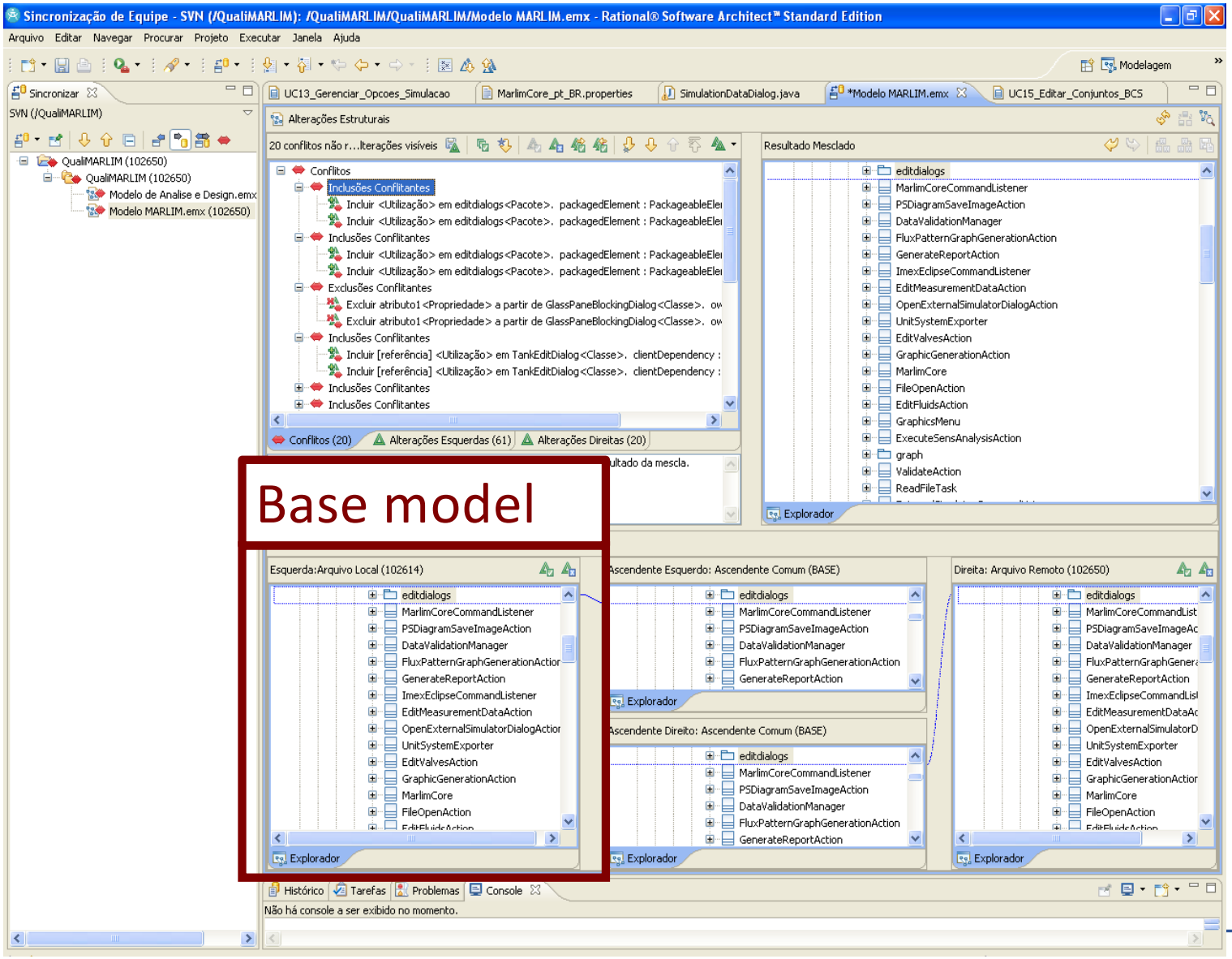
**How do software developers merge UML design models in practice?**

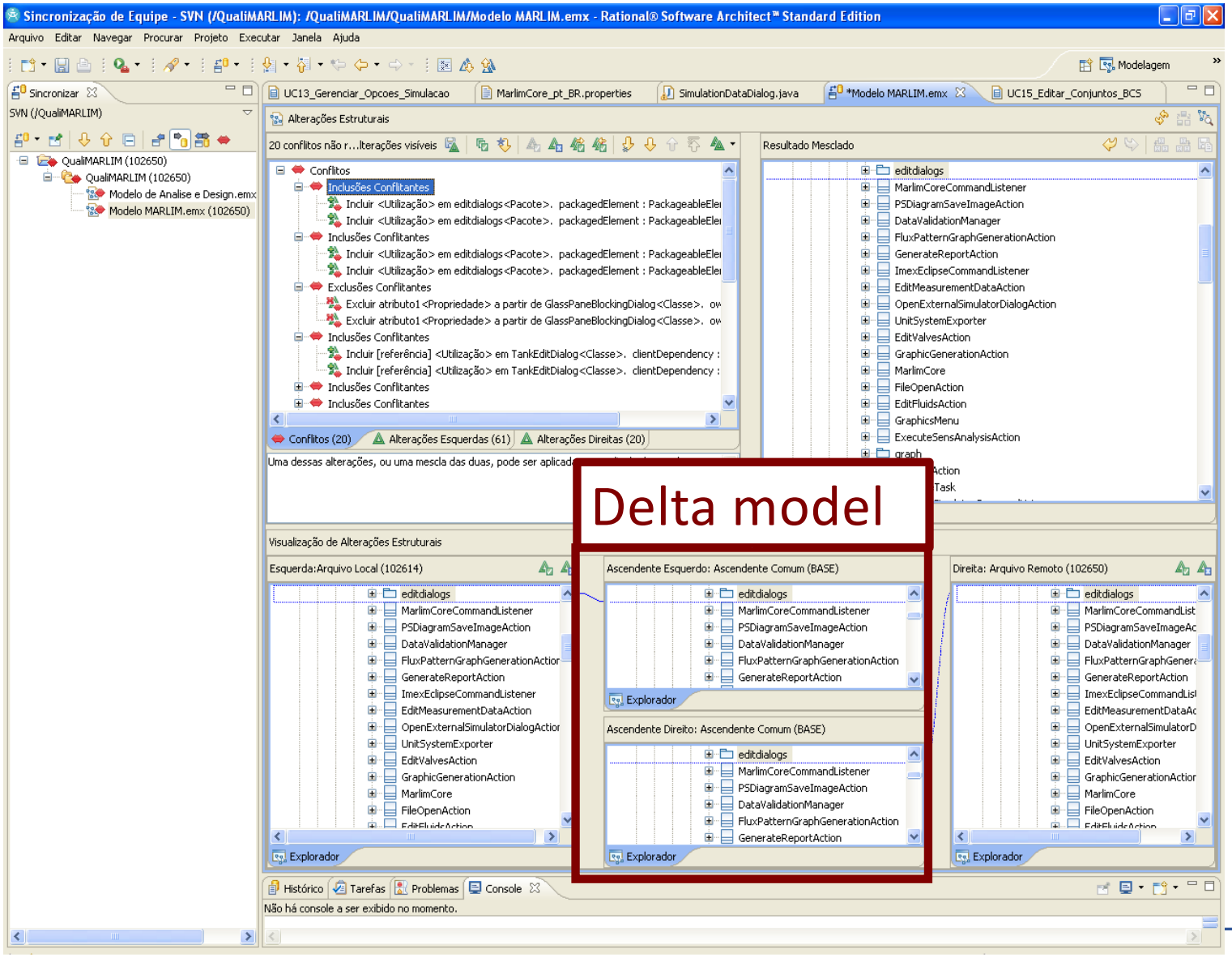
# Study Methodology

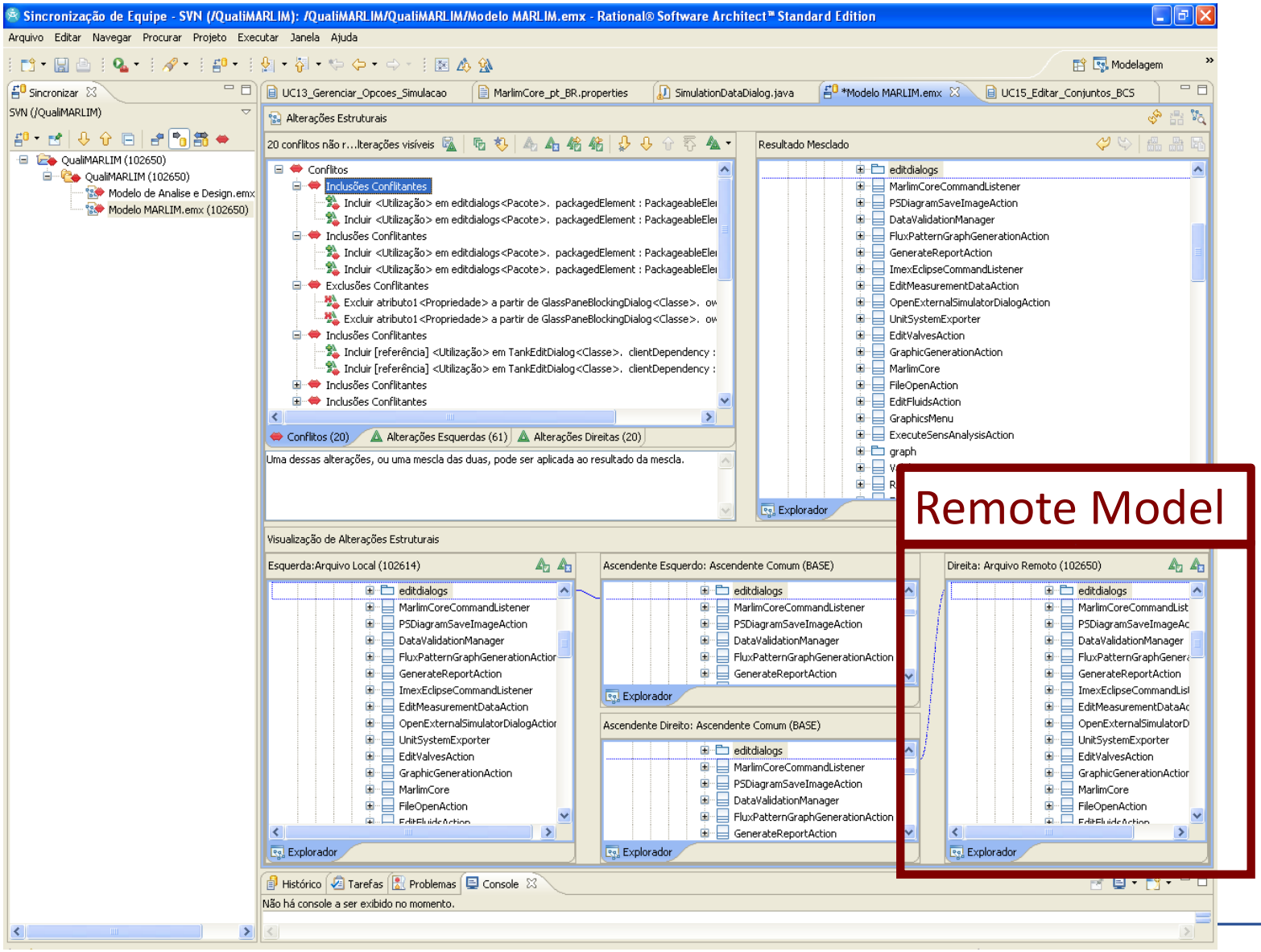




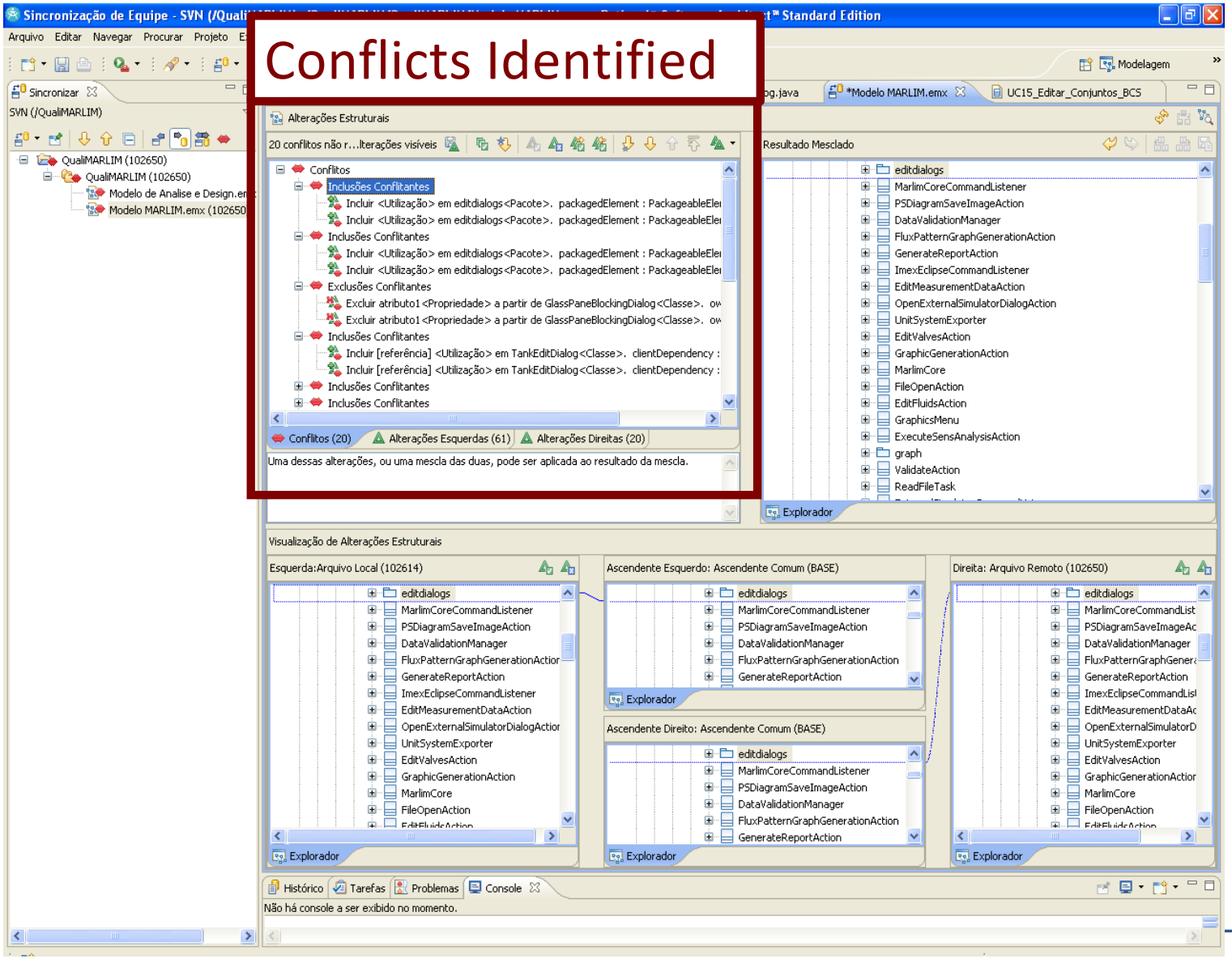

# Merging UML design models using the IBM Rational Software Architect

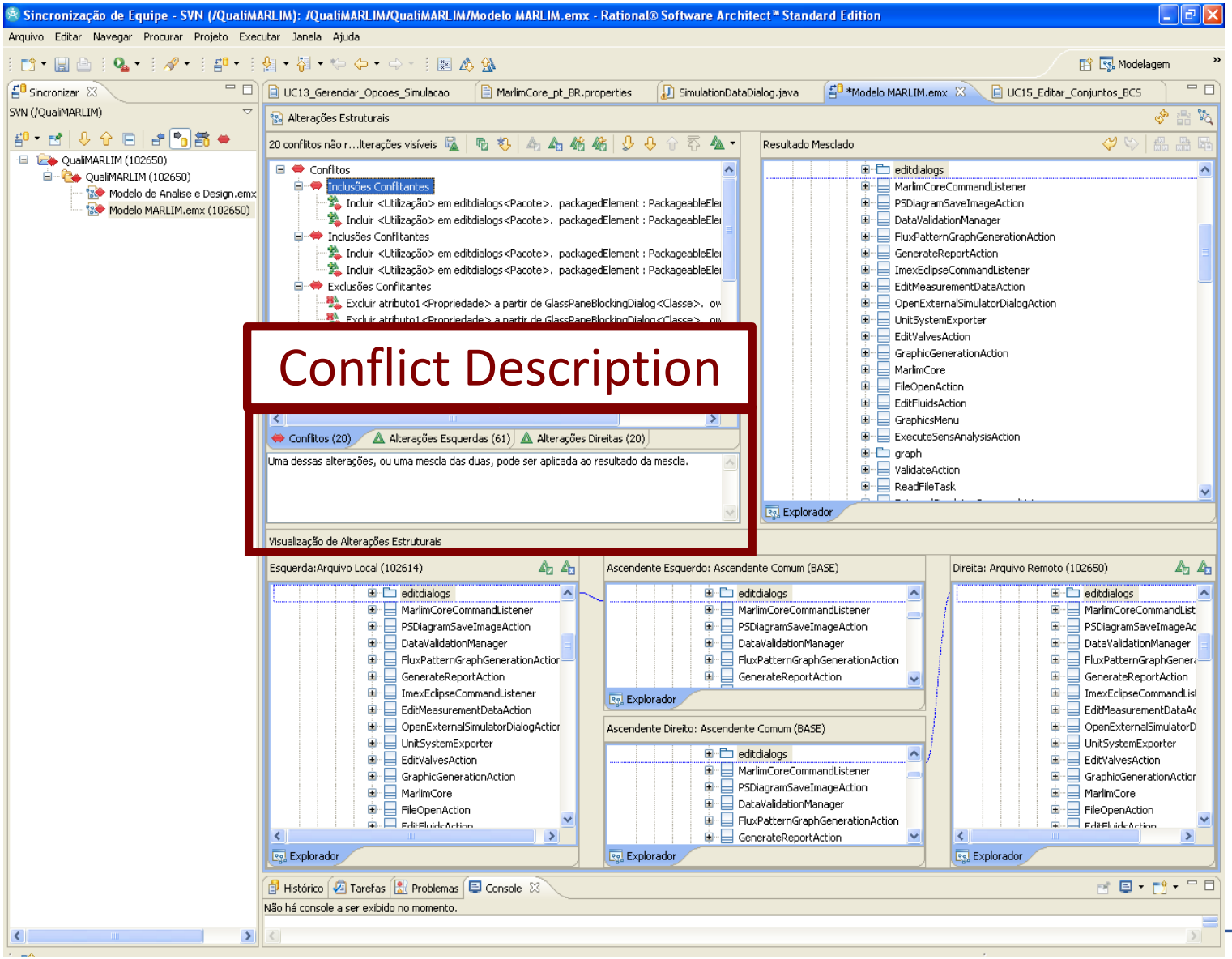

Sincronização de Equipe - SVN (/QualiMARLIM): /QualiMARLIM/QualiMARLIM/Modelo MARLIM.emx - Rational® Software Architect™ Standard Edition
Arquivo Editar Navegar Procurar Projeto Executar Janela Ajuda
Modelagem
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QualiMARLIM (102650)
Modelo de Analise e Design.emx
Modelo MARLIM.emx (102650)
UC13_Gerenciar_Opcoes_Simulacao
MarlimCore_pt_BR.properties
SimulationDataDialog.java
*Modelo MARLIM.emx
UC15_Editar_Conjuntos_BCS
Alterações Estruturais
20 conflitos não r...lterações visíveis
Conflitos
Inclusões Conflitantes
Incluir <Utilização> em editdialogs<Pacote>. packagedElement : PackageableEle
Inclusões Conflitantes
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Excluir atributo1 <Propriedade> a partir de GlassPaneBlockingDialog<Classe>. ow
Inclusões Conflitantes
Incluir [referência] <Utilização> em TankEditDialog<Classe>. clientDependency :
Inclusões Conflitantes
Inclusões Conflitantes
Conflitos (20)
Alterações Esquerdas (61)
Alterações Direitas (20)
Resultado Mesclado
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DataValidationManager
FluxPatternGraphGenerationAction
GenerateReportAction
ImexEclipseCommandListener
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UnitSystemExporter
EditValvesAction
GraphicGenerationAction
MarlimCore
FileOpenAction
EditFluidsAction
GraphicsMenu
ExecuteSensAnalysisAction
graph
ValidateAction
ReadFileTask
Explorador
Base model
Esquerda:Arquivo Local (102614)
Ascendente Esquerdo: Ascendente Comum (BASE)
Ascendente Direito: Ascendente Comum (BASE)
Direita: Arquivo Remoto (102650)
Histórico
Tarefas
Problemas
Console
Não há console a ser exibido no momento.


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GraphicsMenu
ExecuteSensAnalysisAction
graph
Action
Task
Delta model
Visualização de Alterações Estruturais
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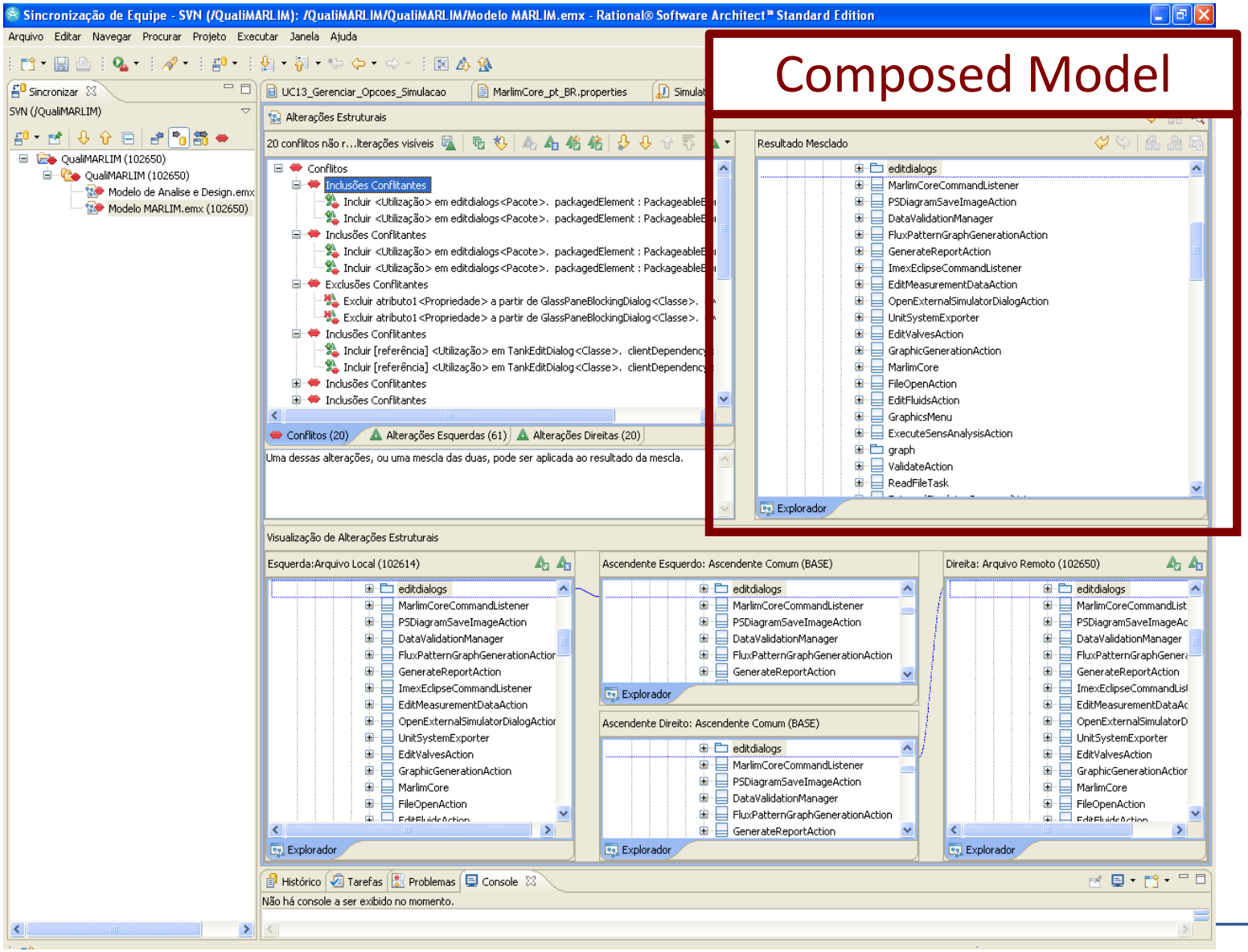

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Console
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Remote Model

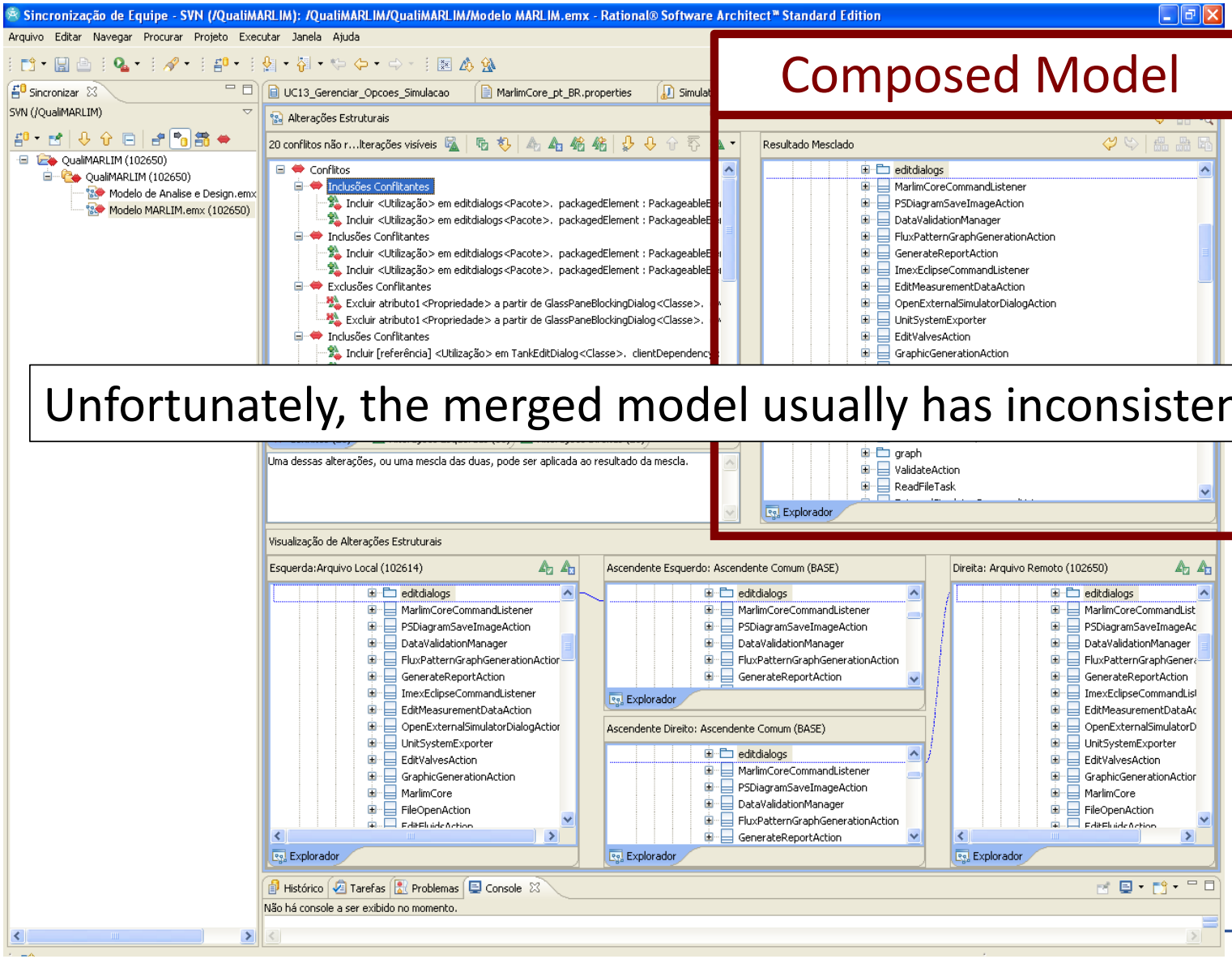
# Conflicts Identified






Conflict Description


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Arquivo Editar Navegar Procurar Projeto Executar Janela Ajuda
Composed Model
Sincronizar
SVN (/QualiMARLIM)
QualiMARLIM (102650)
Modelo de Analise e Design.emx
Modelo MARLIM.emx (102650)
UC13_Gerenciar_Opcoes_Simulacao
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Histórico
Tarefas
Problemas
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# Composed Model

Unfortunately, the merged model usually has inconsistencies or problems.

# Study Methodology

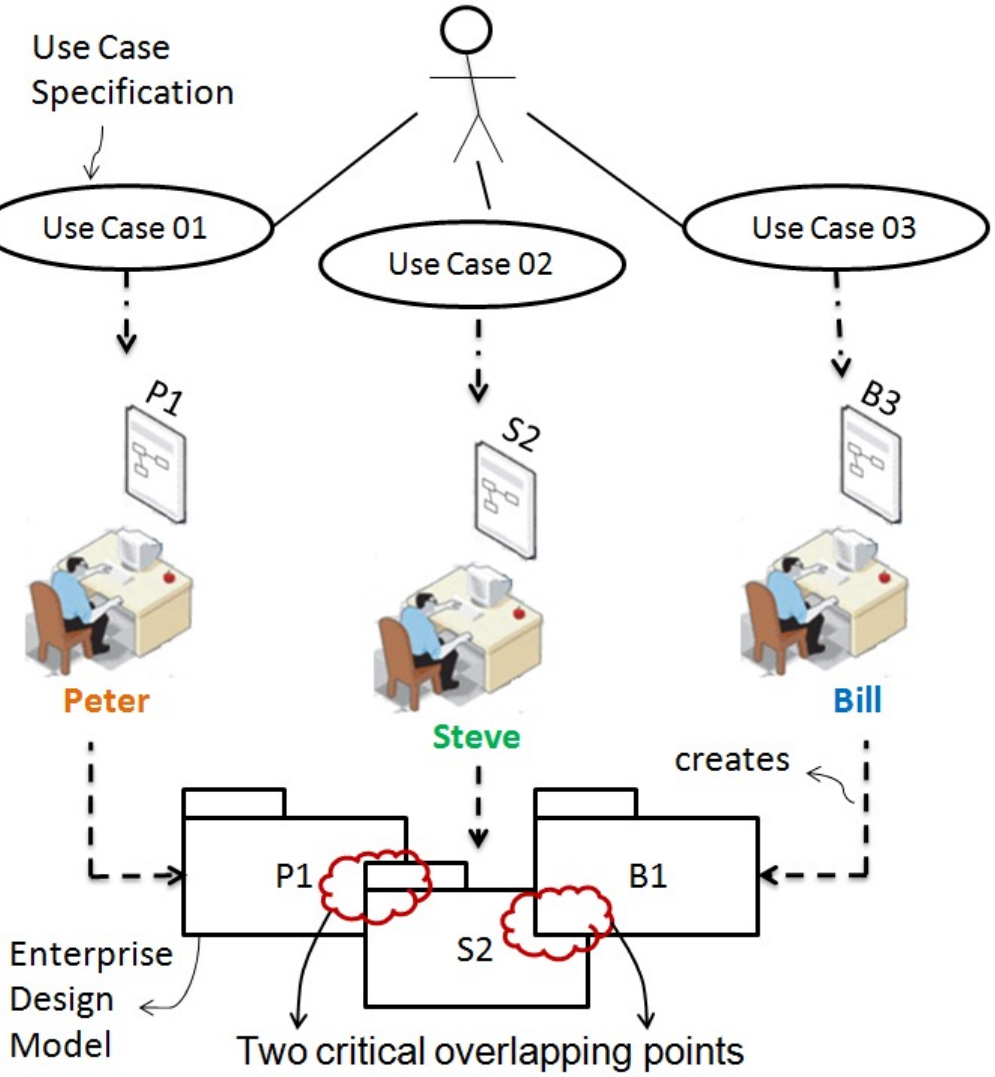
- **Main Goals:**
  1. Understand the effort that software developers invest to merge UML design models
  2. Identify key factors that can affect the composition effort
- **Research Question (RQ):**

**RQ1:** What is the effort to compose UML design models?

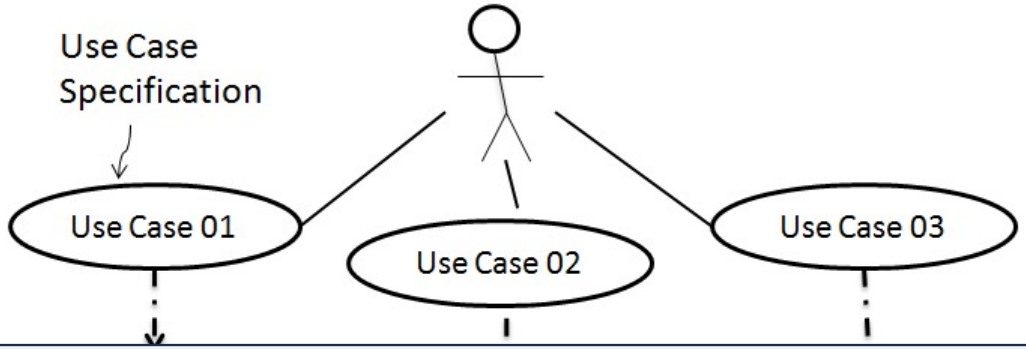
**RQ2:** What are the main factors that affect the composition effort?

# Industrial Case Study

Models are created collaboratively based on use case specifications. However, these models typically have overlapping points, leading to conflicts.

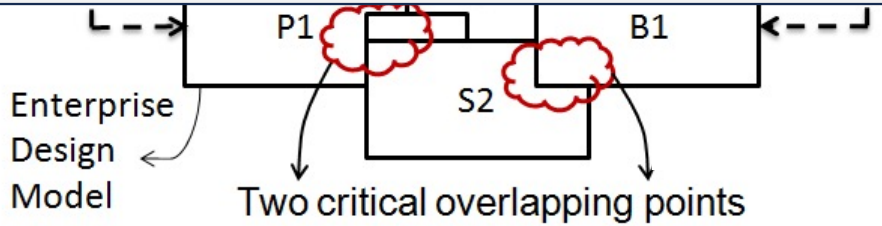


# Industrial Case Study




**How do software developers determine what the integrated UML design model should contain?**

Based on the information collected from the use case specifications



# Study Methodology

- **Study Process**

  **Phase 1: Creation and Composition of the UML design models**

  1. Developers created the design models based on use case specifications, and after composed them using the IBM RSA
  2. Compositions were recorded to enable the quantification of effort

  **Phase 2: Review of the design models created**

  1. Developers detected and resolved the inconsistencies in the output composed model
  2. Developers manually resolved the inconsistencies

  **Phase 3: Interview and Questionnaire**

  1. Developers were interviewed to clarify the collected data
  2. Additional qualitative data were collected using a questionnaire

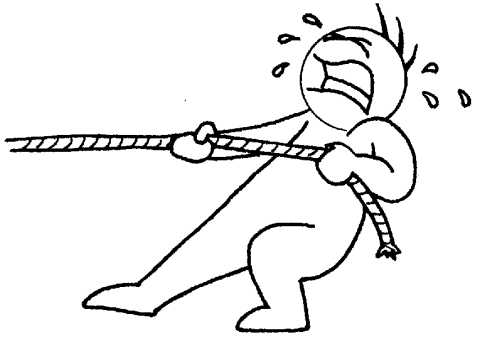
# Composition Effort in Practice?

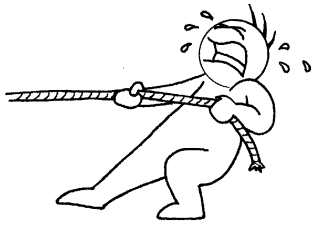
## Our Intuition

Increase up-front work ↓

Minimize lots of “fixes” ↓ ↓

| Application Effort | | Detection Effort | | Resolution Effort |
|---|---|---|---|---|
| High | > | Low | > | Low |







# What … are the main results?

# Quantifying the composition effort

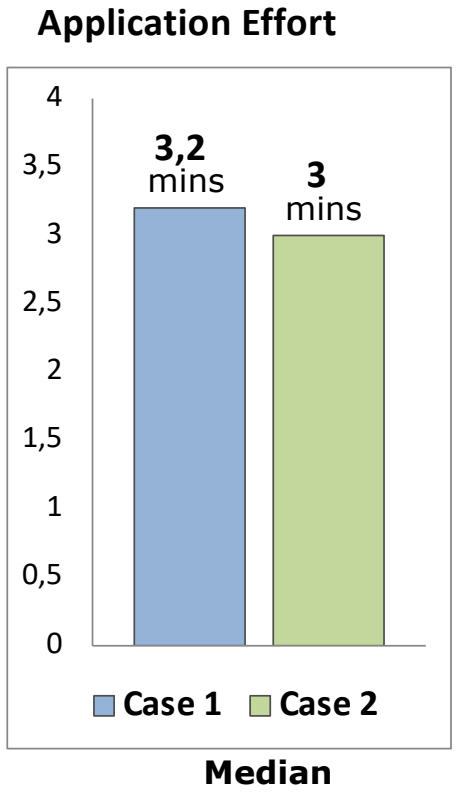
# Composition Effort Analysis

**RQ1:** What is the effort to compose UML design models?

## Application Effort was LOW:

**3.1** mins

The effort to apply semi-automated composition technique is low even for design models of large-scale software




## Intuition not confirmed:

The result contradicts somehow that the semi-automated technique requires a high effort to compose or merge UML design models

# Composition Effort Analysis

## Results:

**Software developers minimize up-front work**

| Application Effort | Detection Effort | Resolution Effort |
|---|---|---|
| Low | ? | ? |



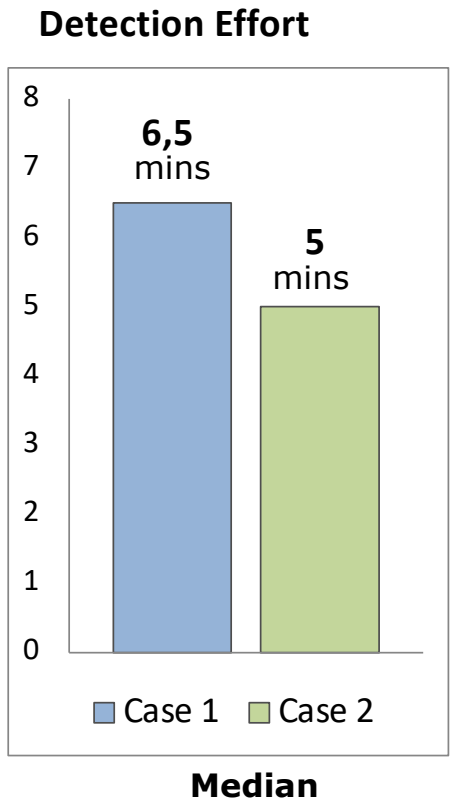
# Composition Effort Analysis

**RQ1:** What is the effort to compose UML design models?

## Detection Effort was "HIGH":

**5.7** mins

The effort to detect inconsistencies using a semi-automated composition technique is relatively high, compared to apply the technique




## Intuition not confirmed:

The result does not confirm that the semi-automated model composition technique requires a low effort to detect inconsistencies.

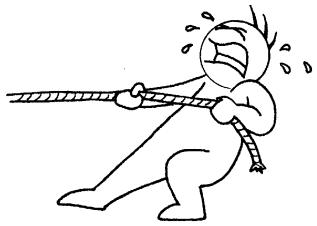
# Composition Effort Analysis

## Results:

**Software developers minimize up-front work**

| Application Effort | Detection Effort | Resolution Effort |
|---|---|---|
| Low | High | ? |





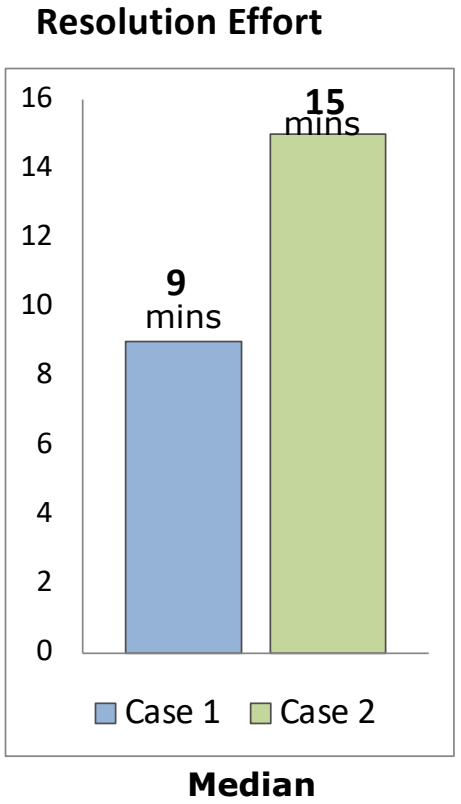
# Composition Effort Analysis

**RQ1:** What is the effort to compose UML design models?

## Resolution Effort was "HIGH":

**12** mins

The effort to resolve inconsistencies in a semi-automated composition technique is high, compared to applying the technique.




## Intuition not confirmed:

The result does not confirm our initial intuition that a semi-automated technique would require a low effort to resolve inconsistencies, compared to the effort to apply the technique.

# Composition Effort Analysis

**RQ1:** What is the effort to compose UML design models?

## Key Finding:

Software developers tend to spend more effort resolving inconsistencies than detecting inconsistenties and applying semi-automated composition technique.

| | Apply | Detect | Resolve |
|---|---|---|---|
| Case Study 1 | 3.2 | 6.5 | 9 |
| Case Study 2 | 3 | 5 | 15 |

≈ **3.1** mins < **5.7** mins < **12** mins

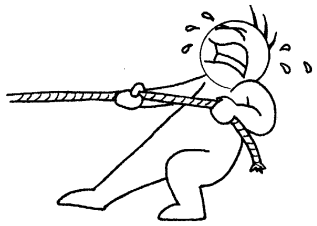
# Composition Effort Analysis

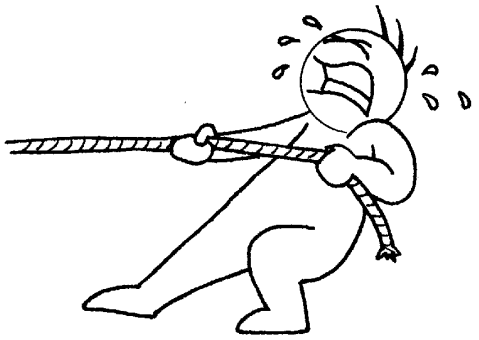
## Results:

**Software developers minimize up-front work**

| Application Effort | Detection Effort | Resolution Effort |
|---|---|---|
| Low | High | High |







# Influential Factors on Composition Effort

**RQ2:** What are the main factors that affect the composition effort?

- **Open-Closed Principle & Effort:**
  1. Changes aligned with the Open–Closed Principle (mostly additions) require lower composition effort.
  2. Changes that violate the Open–Closed Principle (mostly modifications and derivations) require higher composition effort.

Open-closed principle: "software should be open for extensions, but closed for modifications"

# Influential Factors on Composition Effort

**RQ2:** What are the main factors that affect the composition effort?

- **The Conflict-Management Problem:**
  1. The problem is not the number of conflicts,
     but the complexity to understand the conflict resolution.
  2. The higher the number of changes affecting different concerns in the delta model,
     the higher the chance of complex conflicts.
  3. The higher the cohesion of the delta model,
     the lower the complexity of the composition conflicts.

# Influential Factors on Composition Effort

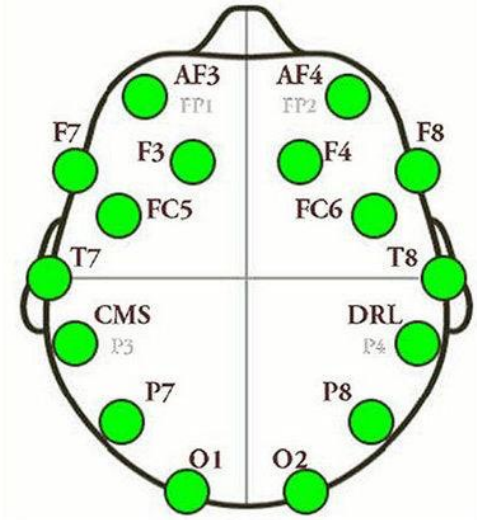
**RQ2:** What are the main factors that affect the composition effort?

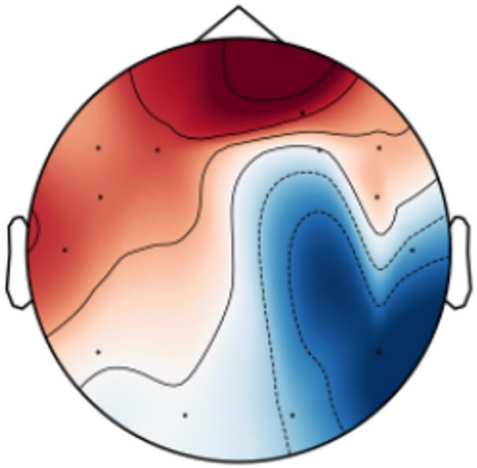
- **The Reputation of Developers Matters:**
  1. Developer reputation influences composition conflict resolution.
  2. Technical reputation has a stronger impact than social reputation.
  3. Less experienced developers tend to accept changes made by more experienced developers.
  4. Changes made by experienced developers tend to remain unchanged over time.

*Understanding what increases composition effort leads to…*

**RQ3:** What happens in developers' brains during software merge?

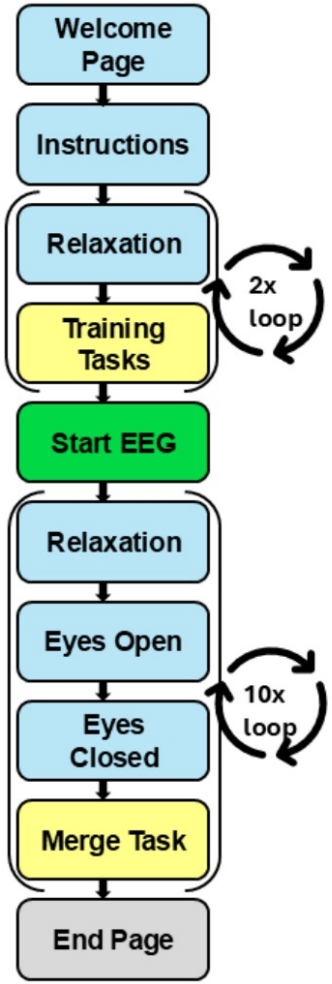


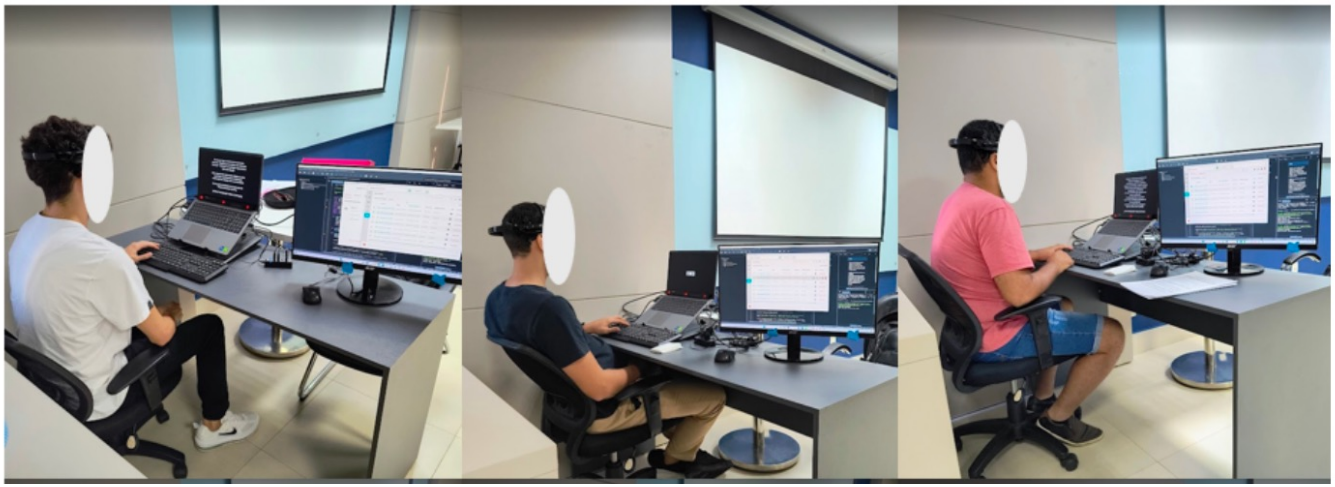


Source: Emotiv

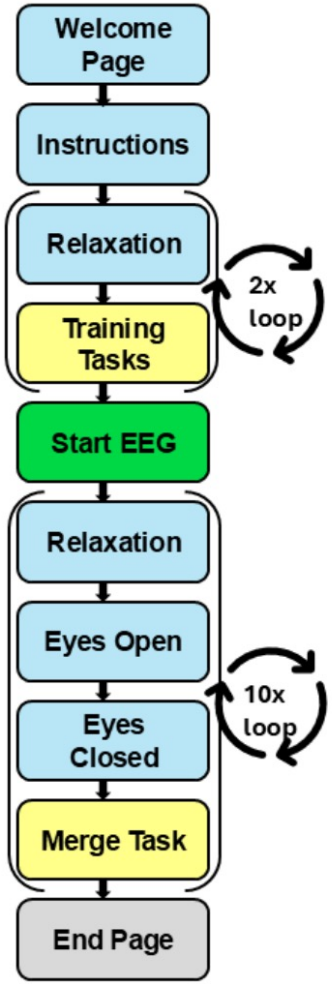
# Controlled Experiment with 42 Participants

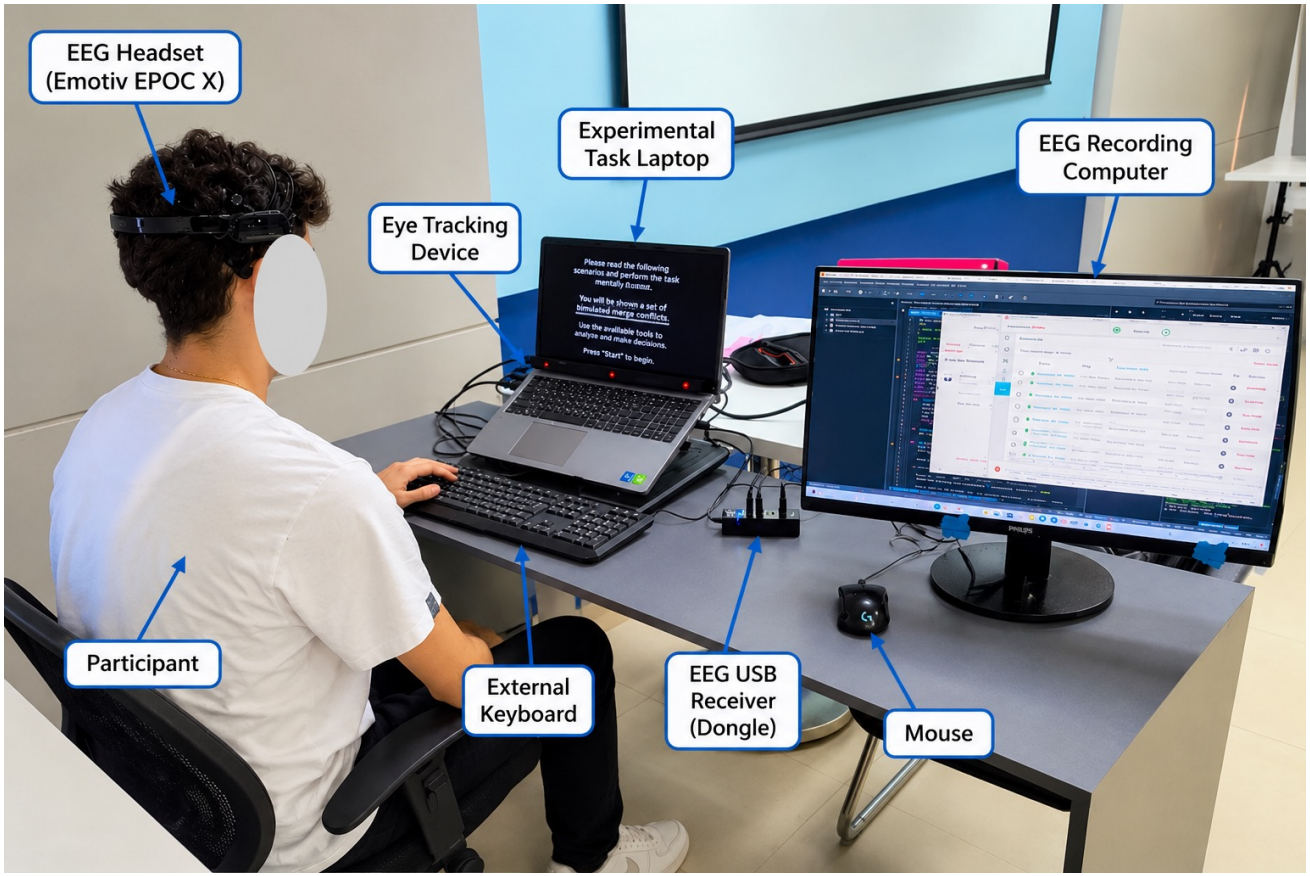


| Attribute | Description |
|---|---|
| Number of subjects | 42 |
| Sessions per subject | 2 (Resting-state, Task-state) |
| Task conditions | Rest, Code Refactoring |
| EEG channels | 14 |
| Sampling rate | 128 Hz |
| Data types | Raw EEG, Preprocessed EEG |
| File format | CSV |
| Data organization | `EEG\, EEGRawData\filtered` |

# Study Setup

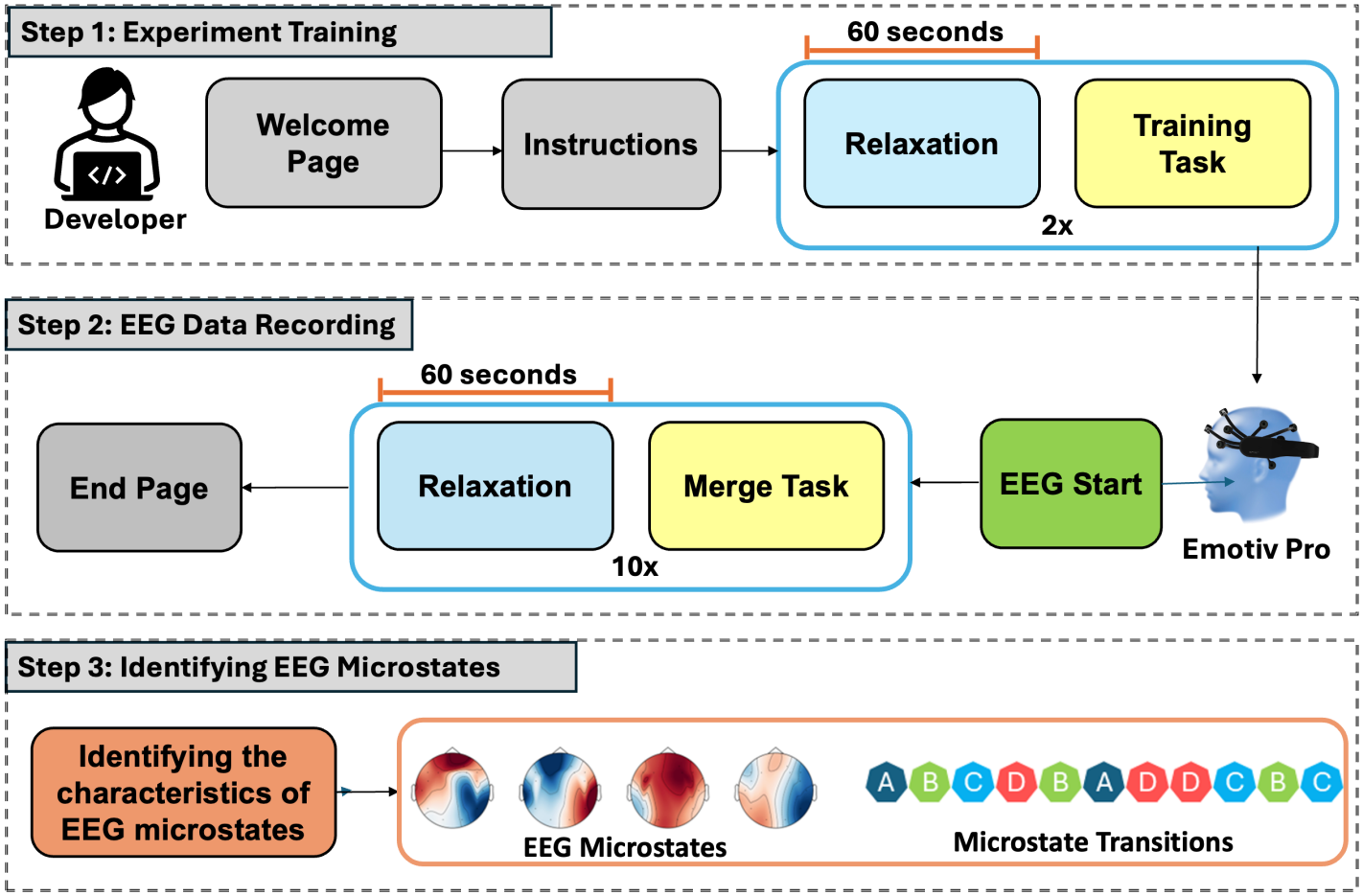



Dataset: https://zenodo.org/records/17386294

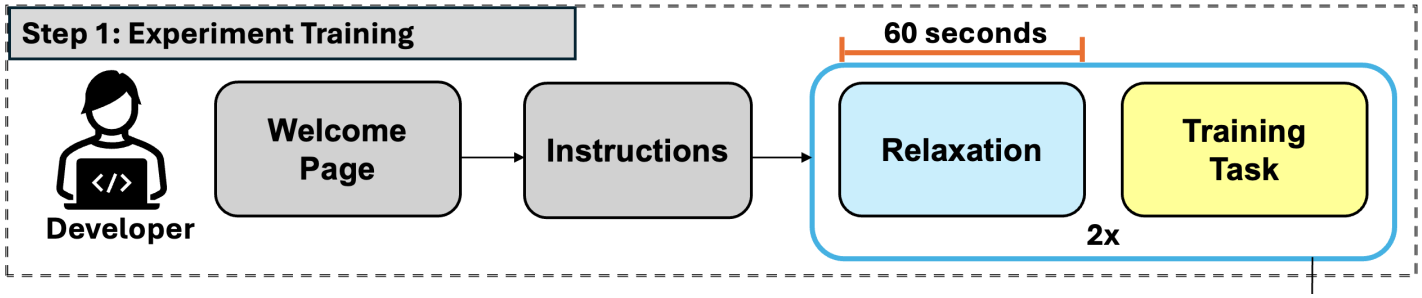
# Study Methodology

## Experimental Process



# Study Methodology

## Experimental Process



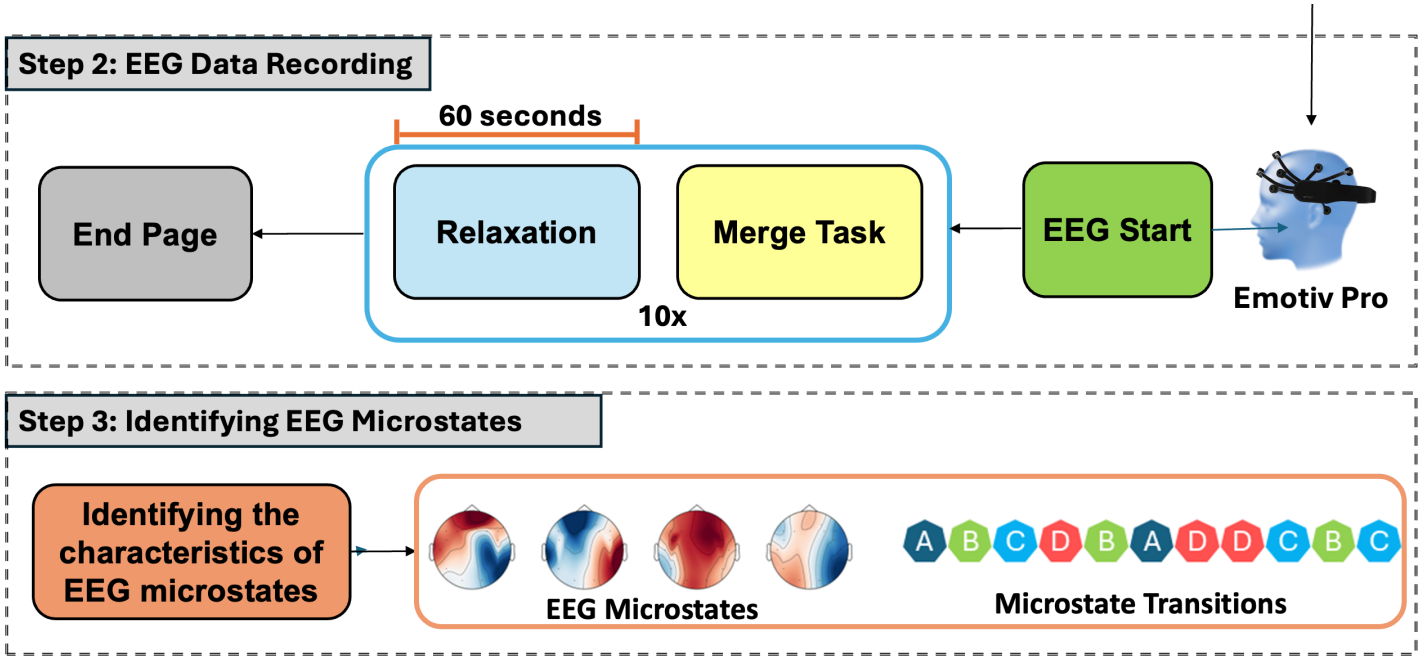

### Phase 1: Participant Preparation

1. Welcome participants and obtain informed consent
2. Collect demographic information
3. Train participants with two practice tasks
4. Calibrate the EEG headset and verify signal quality

# Study Methodology

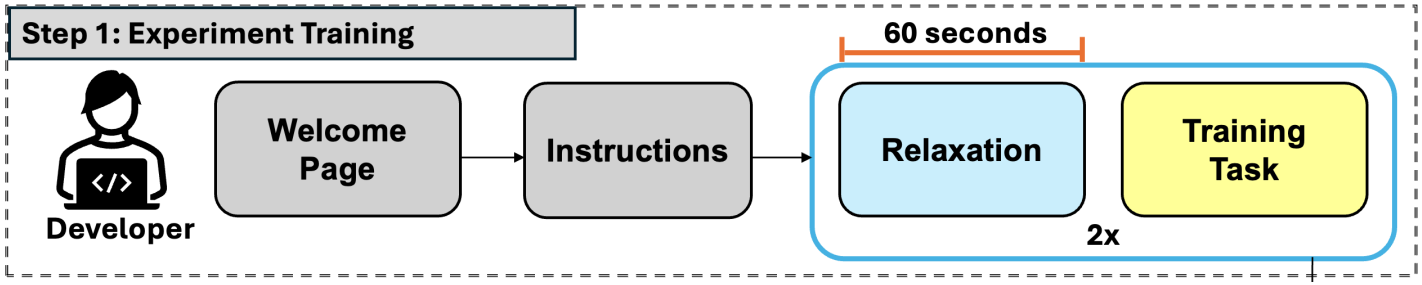
## Phase 2: EEG Data Collection

1. Record a 60-second resting-state baseline
2. Present 10 randomized merge/refactoring tasks
3. Continuously record EEG during task execution
4. Synchronize EEG records with experimental events + eye tracking



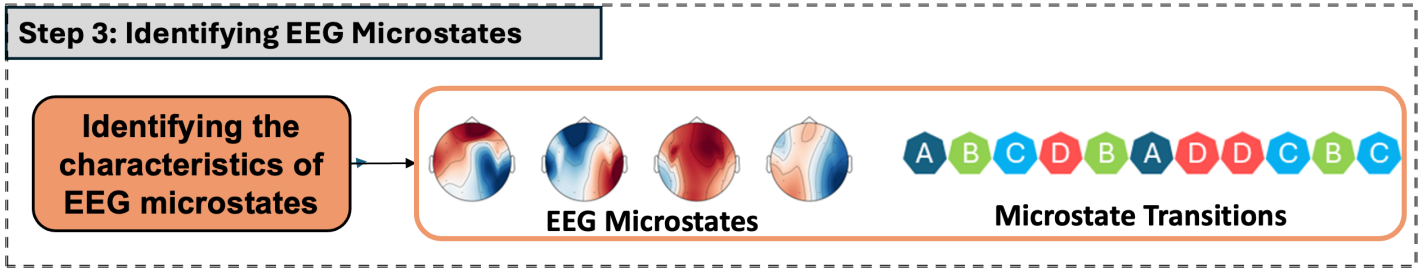
# Study Methodology

## Experimental Process



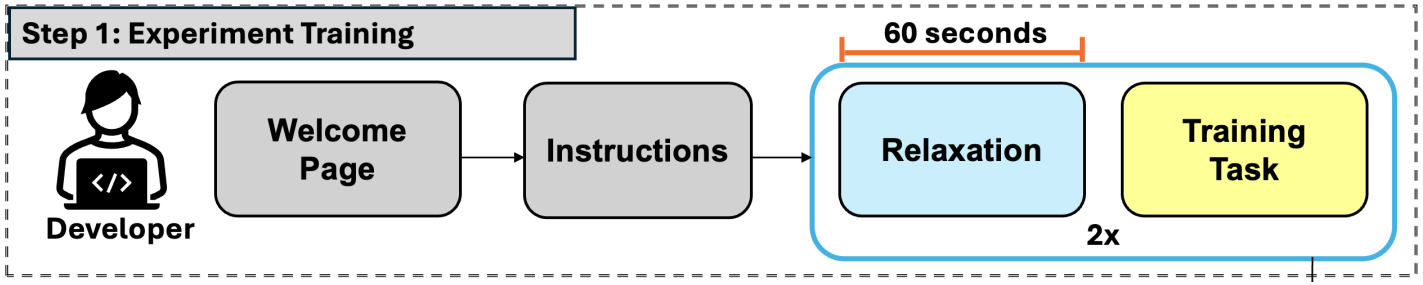

## Phase 3: EEG Processing

1. Remove noisy channels and physiological artifacts (ICA)
2. Apply band-pass filtering
3. Extract EEG microstates (A–D)
4. Compute microstate metrics (duration, occurrence, coverage, GFP)



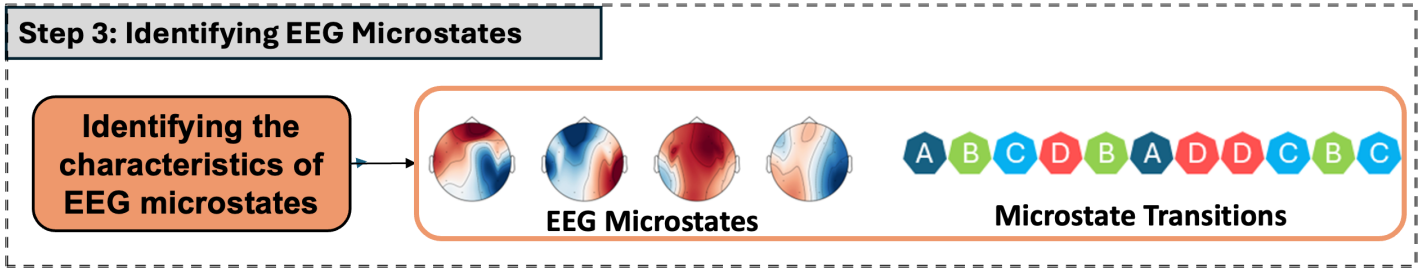
# Study Methodology

## Experimental Process

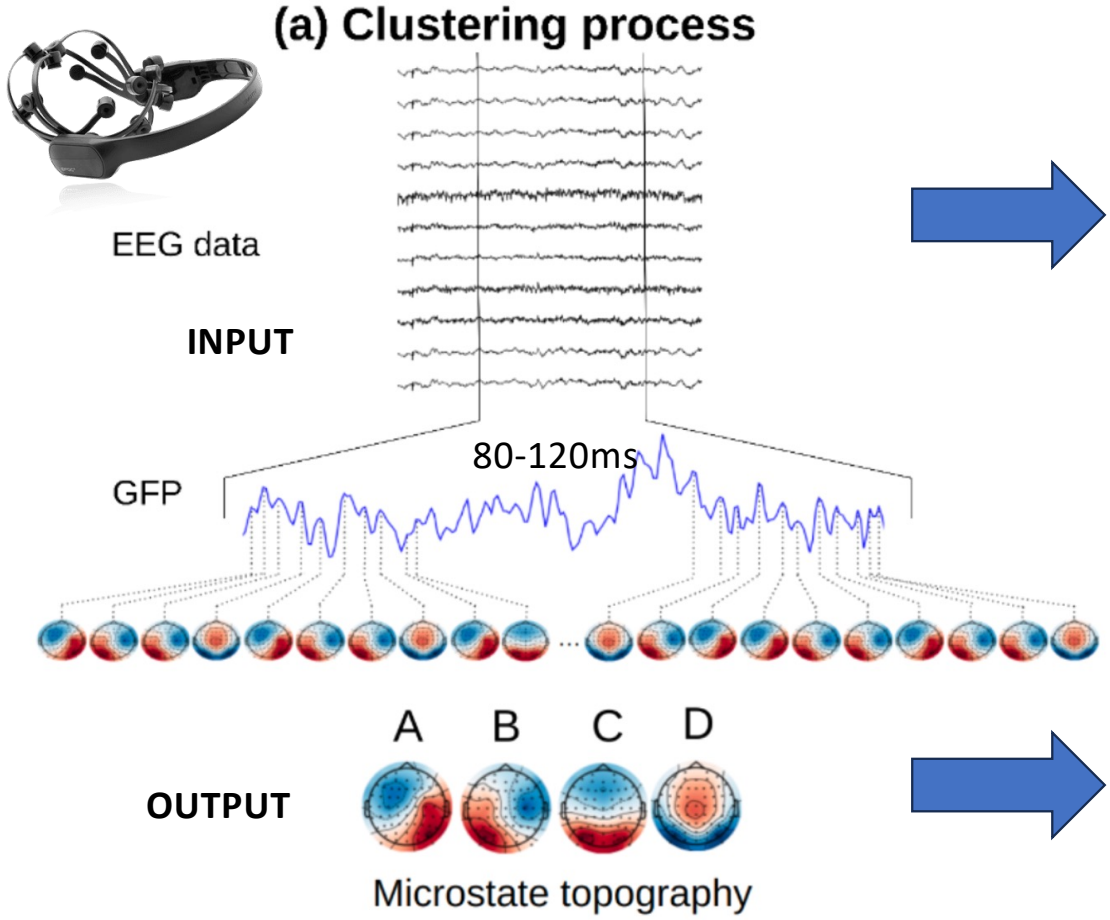



## Phase 4: Data Analysis

1. Compare EEG microstates across task conditions
2. Relate microstate dynamics to task performance
3. Build the **BrainMergeEEG** open dataset
4. Perform statistical analyses and interpret findings



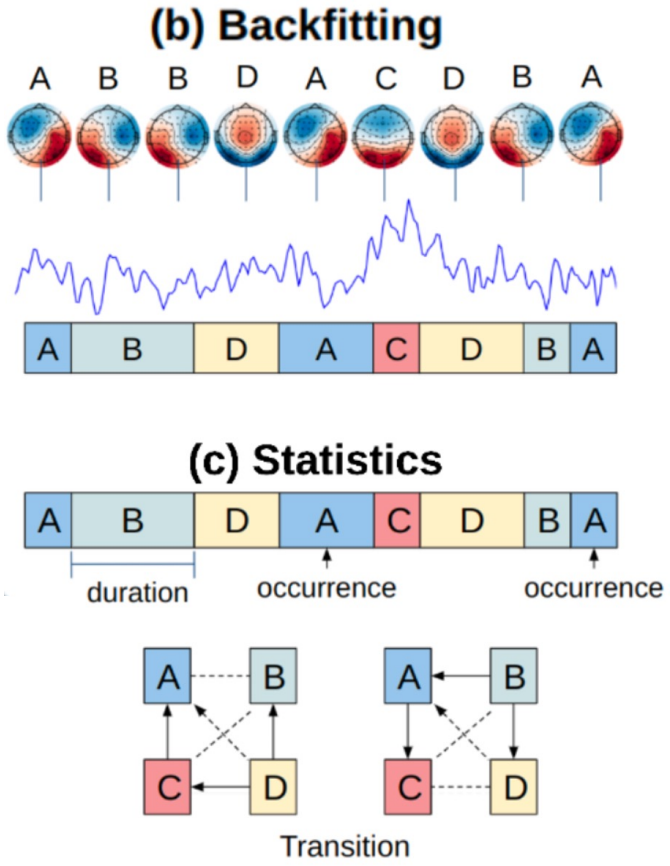
# Study Methodology – EEG Microstates




EEG microstates are obtained by clustering similar EEG topographies into a small set of representative brain states.

EEG microstates are short-lasting, stable patterns of brain activity that represent the basic building blocks of human cognition.

# Study Methodology – EEG Microstates



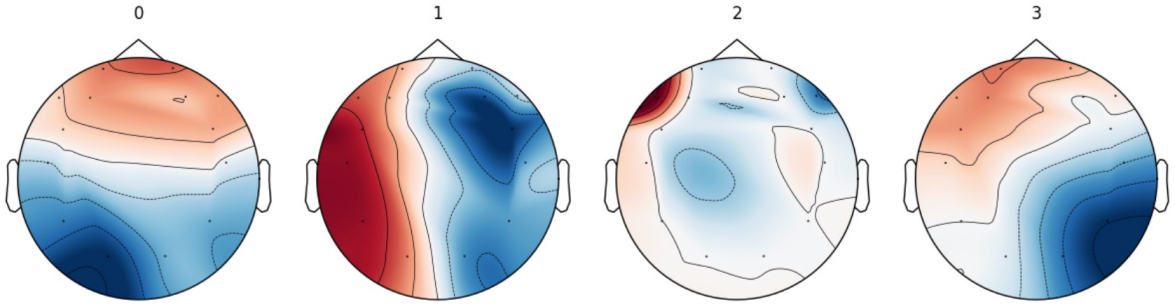

Raw EEG signals are first assigned to the closest canonical microstate (A–D), producing a continuous sequence of brain states over time, transforming continuous EEG recordings into a discrete temporal sequence of microstates.

We compute statistical measures such as duration, occurrence, and transition probabilities between microstates to summarize the temporal dynamics of brain activity.

# Study Methodology

Files

1_ProcessingFiltersEEG.ipynb




```
Channels used: ['AF3', 'F7', 'F3', 'FC5', 'T7', 'P7', 'O1', 'O2', 'P8', 'T8', 'FC6', 'F
4', 'F8', 'AF4']
```

In [ ]:

```
# ———————————————— auxiliary functions ————————————————
import numpy as np, pandas as pd, mne
from pathlib import Path
from collections import Counter

num2str  = np.array(list("ABCD"))       # 0→A, 1→B, …
str2num  = {c: i for i, c in enumerate("ABCD")}   # A→0, B→1, …

pairs_of_interest = ['AB' 'AC' 'AD'
```

SCAN ME

Important:
The dataset and the complete preprocessing pipeline are publicly available on Zenodo.

Dataset: https://zenodo.org/records/17386294

# Study Methodology

## Base Source Code A

```java
//Code A
class Entity {
    protected String name;
    public Entity(String name) { this.name = name; }
    public String getname() { return name; }
}
```

+

## Base Source Code B

```java
//Code B
class BookAuthor {
    private Book Book;
    private Author Author;
    public BookAuthor(Book Book, Author Author) {
        this.Book = Book;
        this.Author = Author;
    }
}
```



## Output Merged Source Code

```java
//Code A+B
class Entity {
    protected String name;
    public Entity(String name) { this.name = name; }
    public String getname() { return name; }
}

class Book extends Entity {
    public Book(String name) { super(name); }
}
class Author extends Entity {
    public Author(String name) { super(name); }
}

class BookAuthor {
    private Book Book;
    private Author Author;
    public BookAuthor(Book Book, Author Author) {
        this.Book = Book;
        this.Author = Author;
    }
}
```

Participants performed 10 source code merge tasks during the experiment.

# Example of Experimental Task - Question



Participants first *read the task description* and then integrate two source code versions according to the specified requirements.

**Task Description In Portuguese**

Cenário 01: O Código A representa o estado atual do projeto de software para Gestão Financeira. O Código B apresenta as modificações que precisam ser inseridas no Código A. As modificações são: adição da função calcularLucro. Como desenvolvedor, você deve gerar uma nova versão do Código A incorporando as alterações presentes no Código B.

**Code A and Code B To Be Merged**

**Code A:**

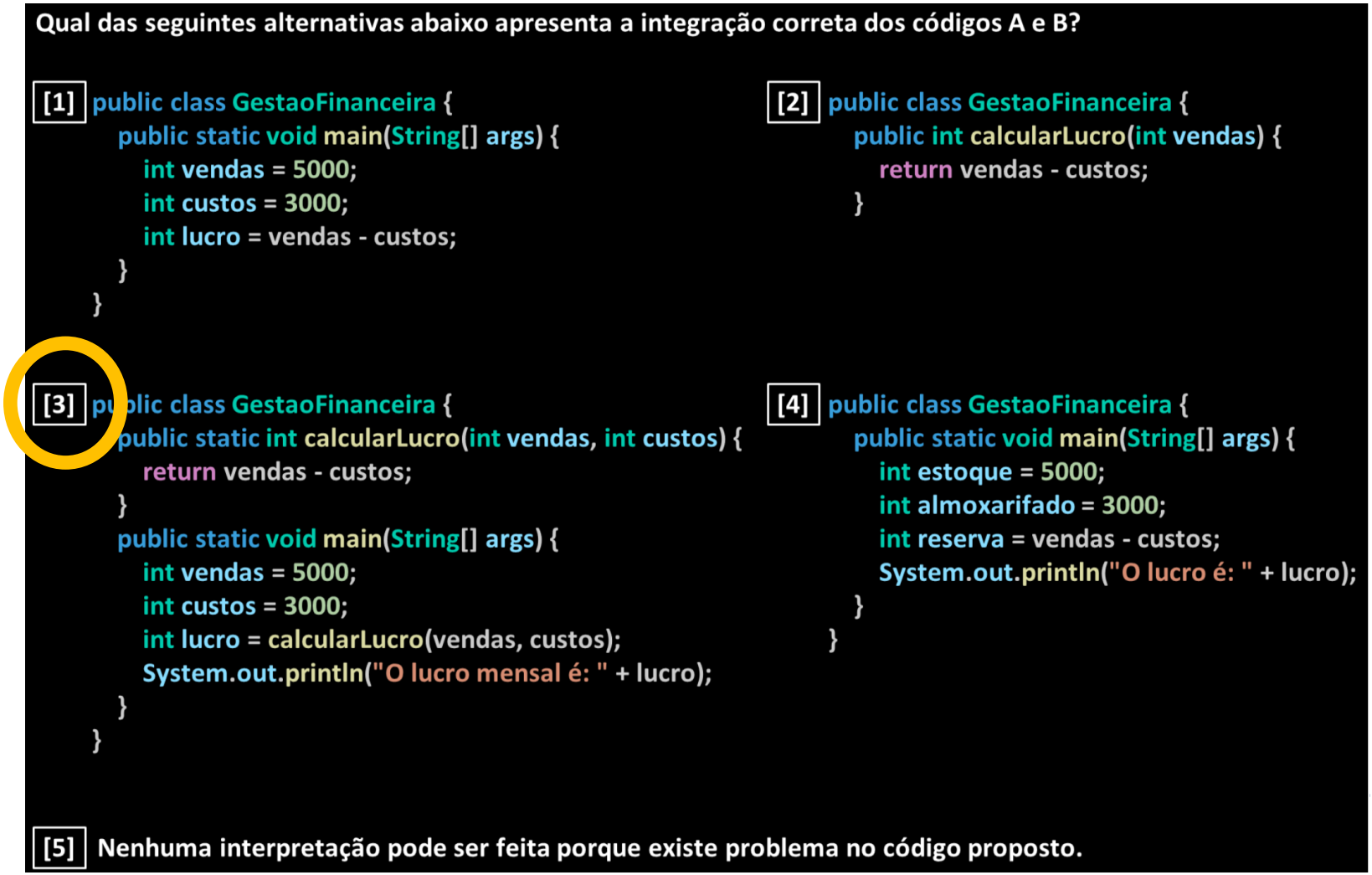
```
public class GestaoFinanceira {
    public static void main(String[] args) {
        int vendas = 5000;
        int custos = 3000;
        int lucro = vendas - custos;
        System.out.println("O lucro é: " + lucro);
    }
}
```

**Code B:**

```
public class GestaoFinanceira {
    public  int calcularLucro(int vendas, int custos) {
        return vendas - custos;
    }
}
```

# Example of Experimental Task - Options

Participants must choose an Alternative, representing the desired merged source code.



# Experimental Outcomes

- **Main Results**

  **BrainMergeEEG Dataset**

  1. First EEG microstate dataset for software merge tasks.
  2. 42 developers performing merge tasks.
  3. 10 randomized merge scenarios.
  4. Open dataset and reproducible pipeline.
  5. Public benchmark for Neuro-Software Engineering.

SCAN ME

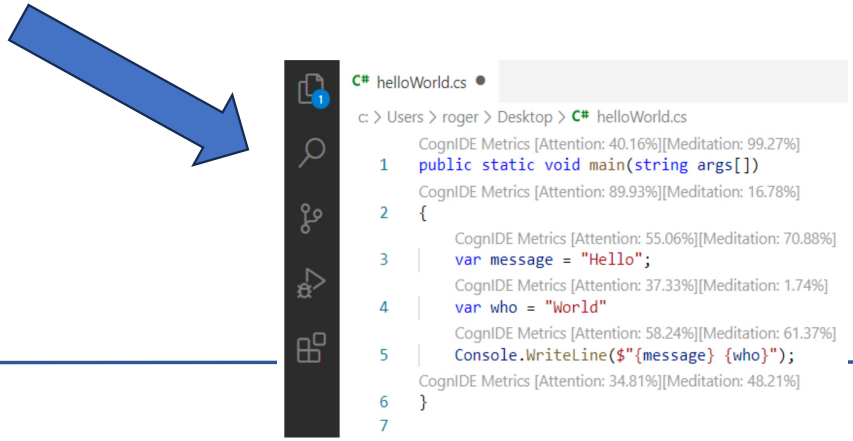


# Experimental Outcomes

- **Main Results**

## Key Findings

1. Distinct EEG microstate patterns emerged during merge tasks.
2. Better performance associated with Microstates A, B, and D.
3. Poorer performance associated with Microstate C.
4. EEG microstates provide objective indicators of developers' cognitive effort.
5. First steps for cognitive-adaptive development environments.



# Experimental Outcomes

- **Main Results**

We have associated biometrics (attention and meditation) to source code in Visual Studio Code.



# Experimental Outcomes

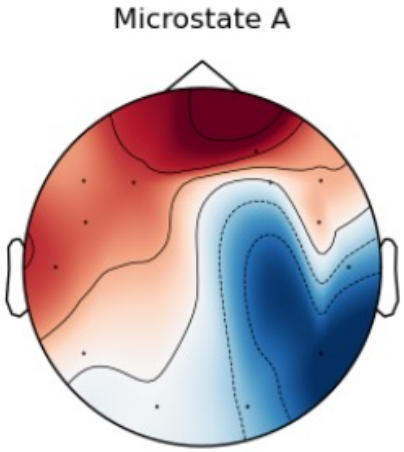
- **P06: Sequence of EEG microstates (A, B, C, D) to perform a merge task**

**Total de microstates: 1.786** (392 (A) + 685 (B) + 233 (C) + 476 (D)).

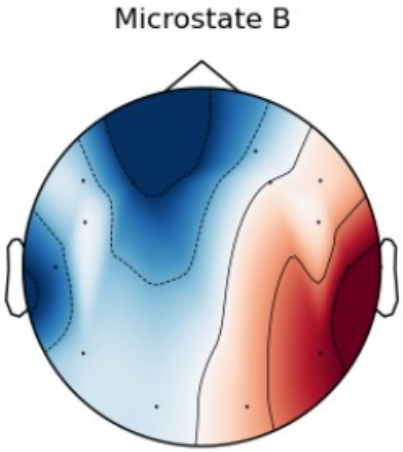

A,A,B,A,B,A,B,A,B,C,B,D,B,A,B,D,D,B,A,D,B,B,B,D,B,B,B,A,D,B,A,C,B,A,A,D,B,B,B,B,B,D,B,C,B,C,A,B,C,B,B,B,A,B,D,D,D,B,B,B,A,A,B,B,C,A,B,D,B,B,D,B,B,D,B,B,
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D,A,B,B,B,A,B,B,A,B,B,B,B,B,B,D,C,B,A,B,B,D,B,A,C,D,D,B,D,A,B,C,B,D,A,A,D,B,B,D,D,C,A,A,D,B,D,B,C,C,D,D,A,D,B,B,D,B,B,B,B,B,A,D,B,A,B,B,A,D,D,B,D,D,B,B,
D,C,B,C,A,A,B,A,D,D,A,D,B,B,D,D,B,C,C,D,A,D,D,D,B,D,B,B,B,C,D,A,C,B,A,B,B,A,A,D,D,D,D,D,C,B,A,D,B,B,C,A,B,D,C,D,D,A,A,A,A,B,B,D,D,D,B,A,B,B,D,D,B,D,D,A
,B,B,D,C,D,B,C,B,D,A,C,B,D,C,C,B,B,D,B,B,D,C,A,A,A,D,A,B,B,B,D,D,B,B,D,A,A,D,B,C,D,A,C,C,D,D,B,D,D,D,B,B,B,B,D,B,C,A,A,D,D,B

# Experimental Outcomes

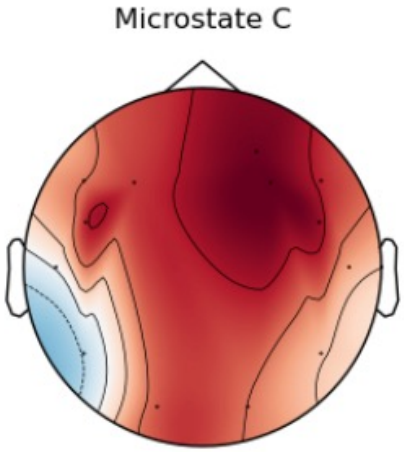
- **Participant 6 (P6): Quantitying and Showing EEG Microstates**



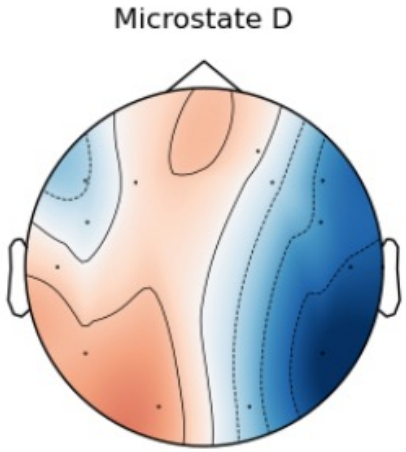

A: 392




B: 685



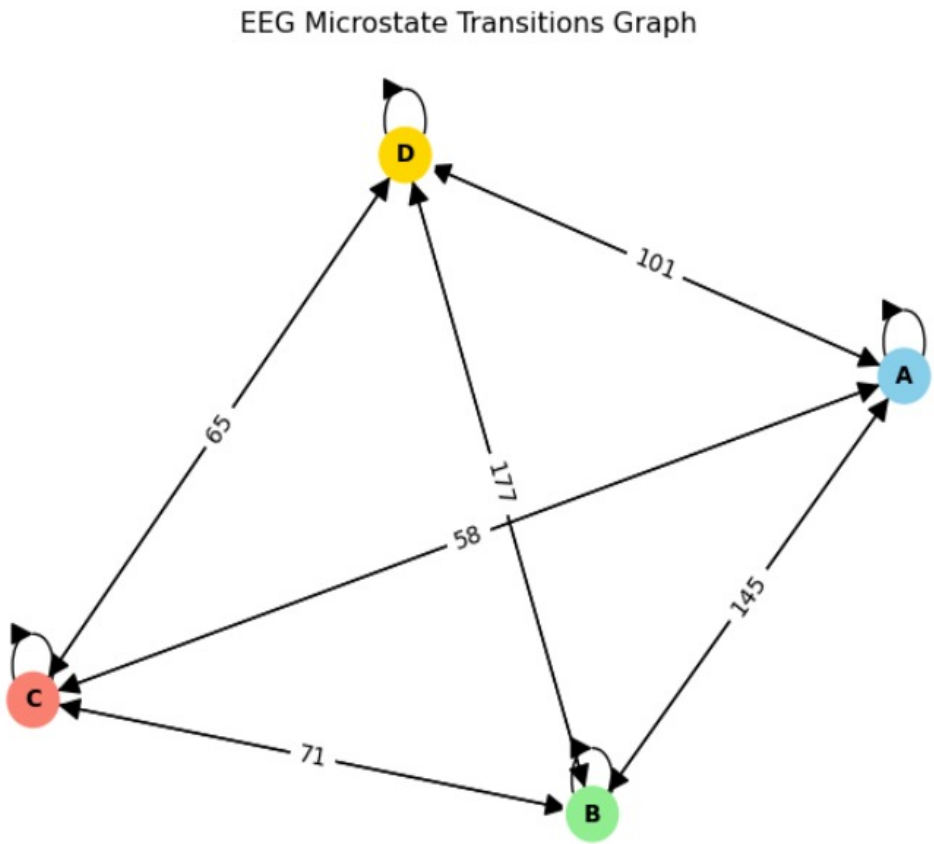

C: 233



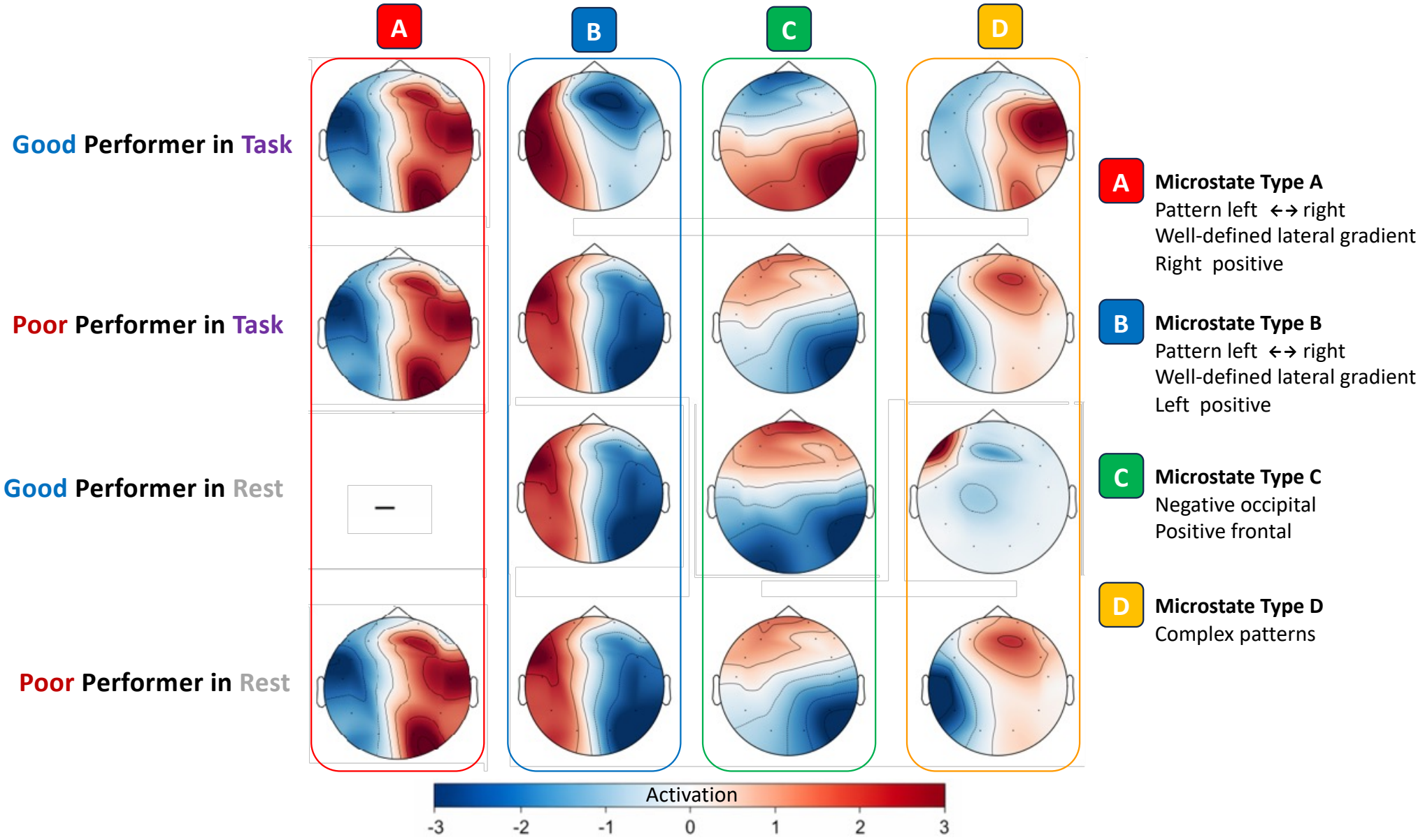

D: 476

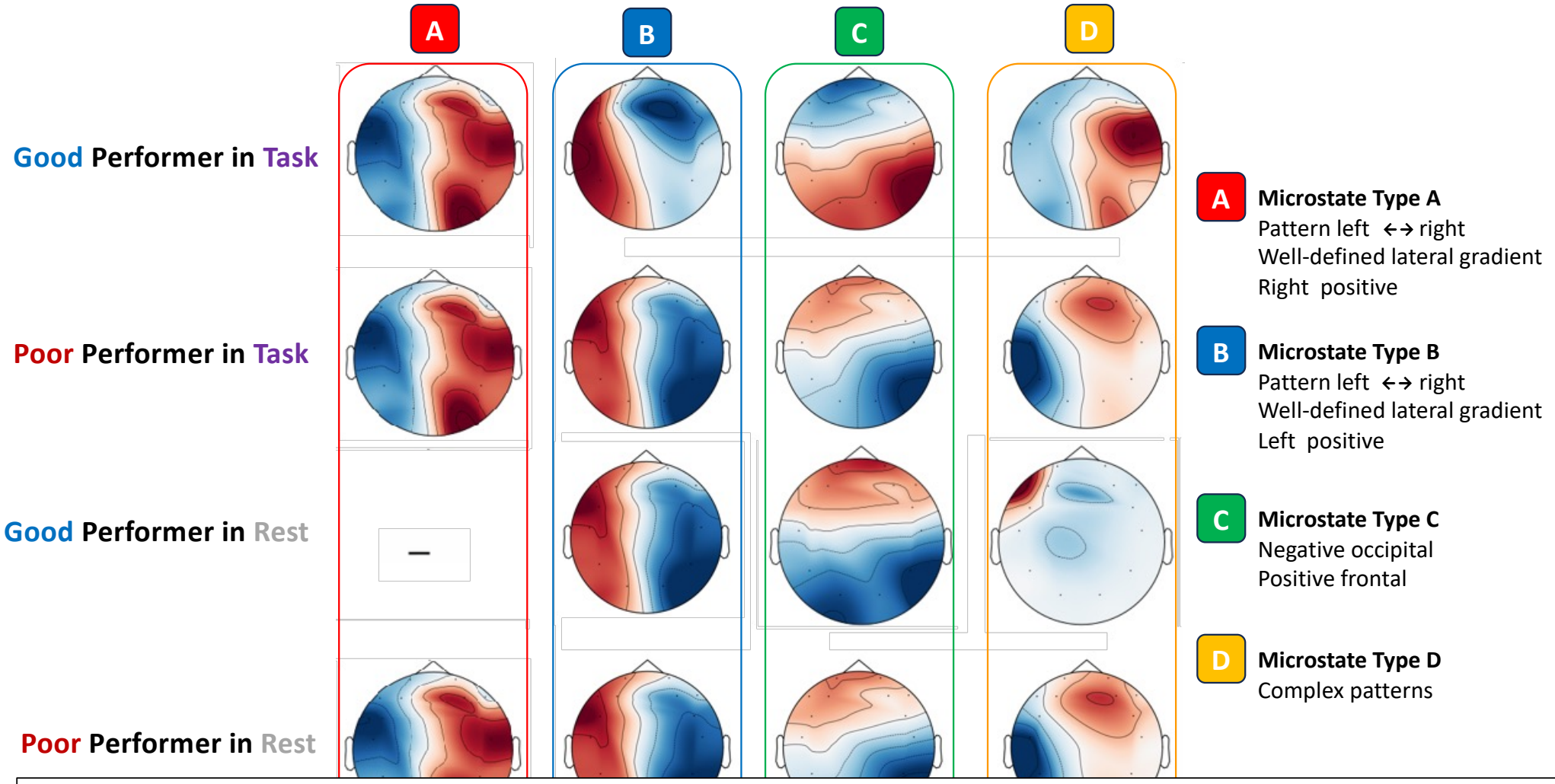
# Experimental Outcomes

- **P06: EEG microstate transitions**




- Each node (A–D) represents an EEG microstate, while each arrow indicates a transition from one brain state to another.
- The numbers on the arrows represent how often each transition occurred during the experiment; higher values indicate more frequent transitions.
- Self-loops indicate that the brain remained in the same microstate before switching, revealing the temporal stability of that cognitive state.


A
B
C
D
Good Performer in Task
Poor Performer in Task
Good Performer in Rest
Poor Performer in Rest
Activation
-3
-2
-1
0
1
2
3
A
Microstate Type A
Pattern left ↔ right
Well-defined lateral gradient
Right positive
B
Microstate Type B
Pattern left ↔ right
Well-defined lateral gradient
Left positive
C
Microstate Type C
Negative occipital
Positive frontal
D
Microstate Type D
Complex patterns




Each column represents one EEG microstate (A–D), while each row corresponds to an experimental condition.

**Good Performer in Task**

*Participants who successfully completed the merge task.*

**Poor Performer in Task**

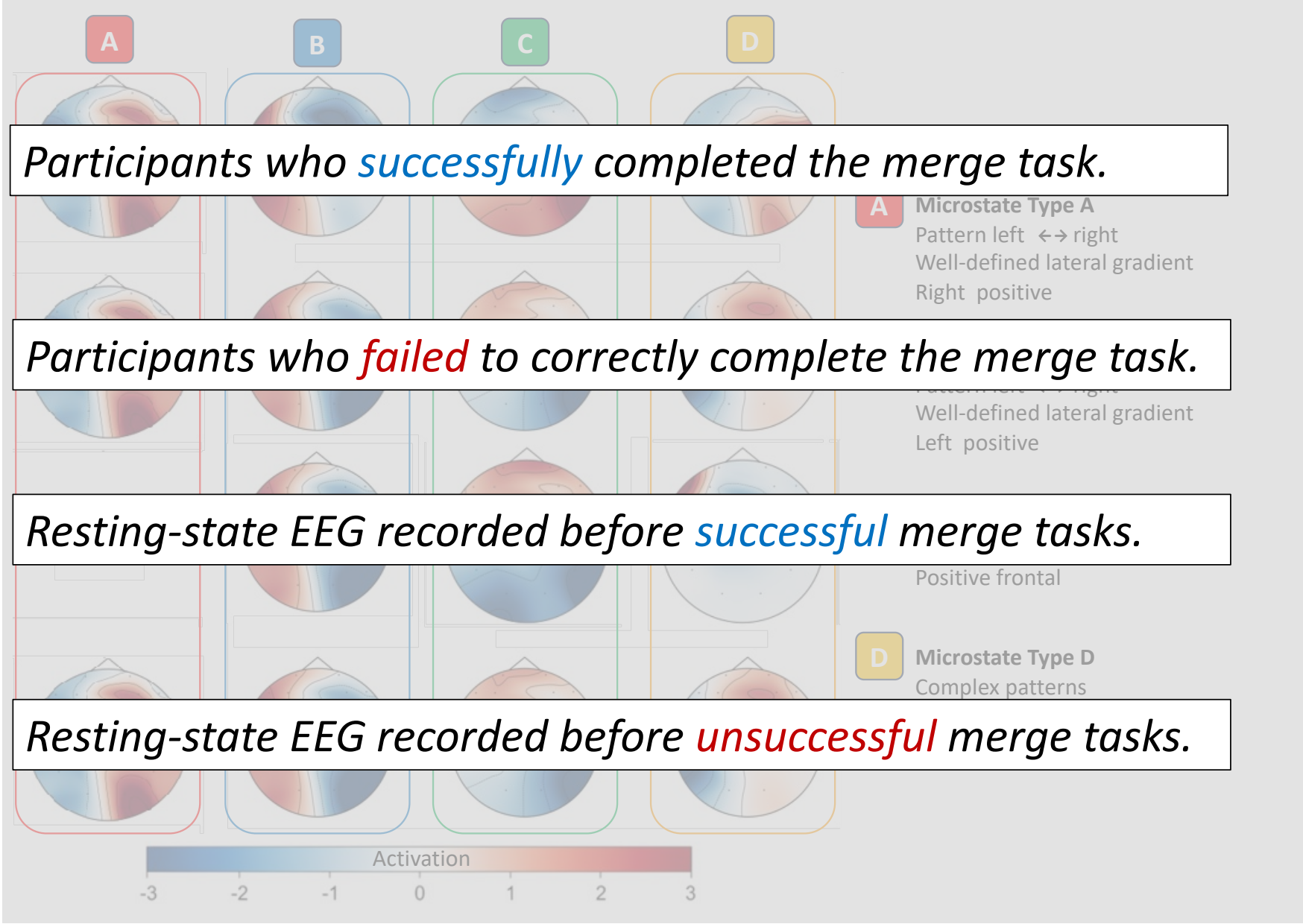
*Participants who failed to correctly complete the merge task.*

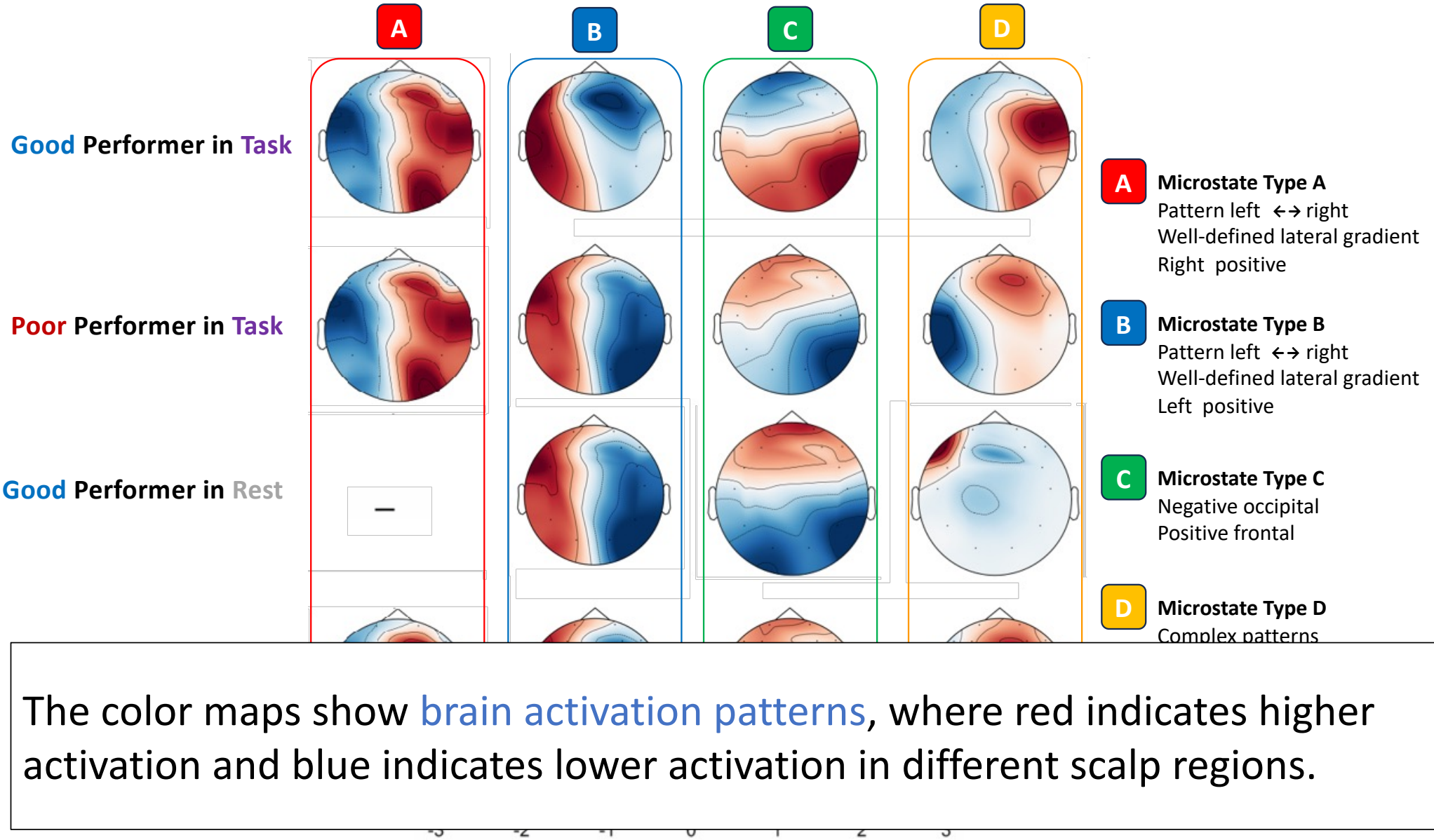
**Good Performer in Rest**

*Resting-state EEG recorded before successful merge tasks.*

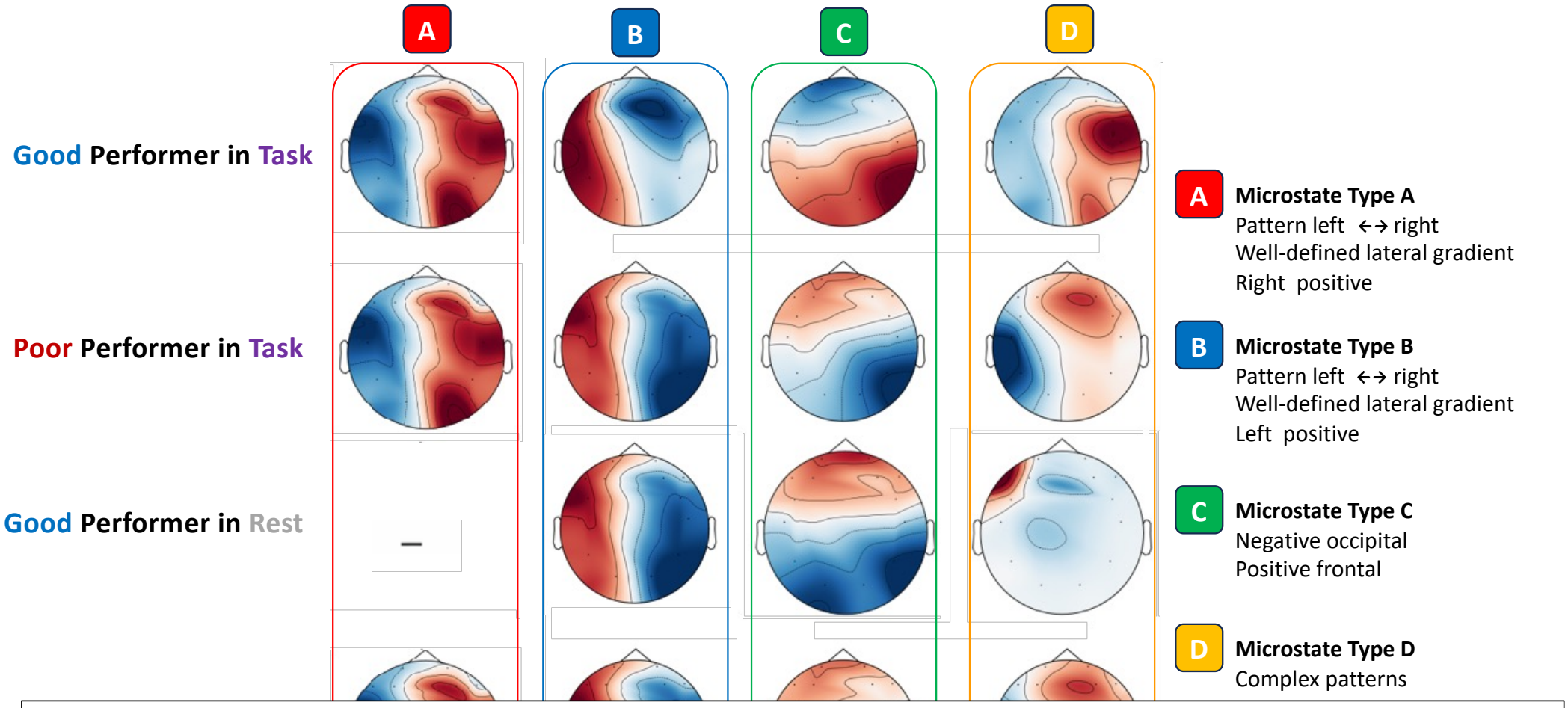
**Poor Performer in Rest**

*Resting-state EEG recorded before unsuccessful merge tasks.*

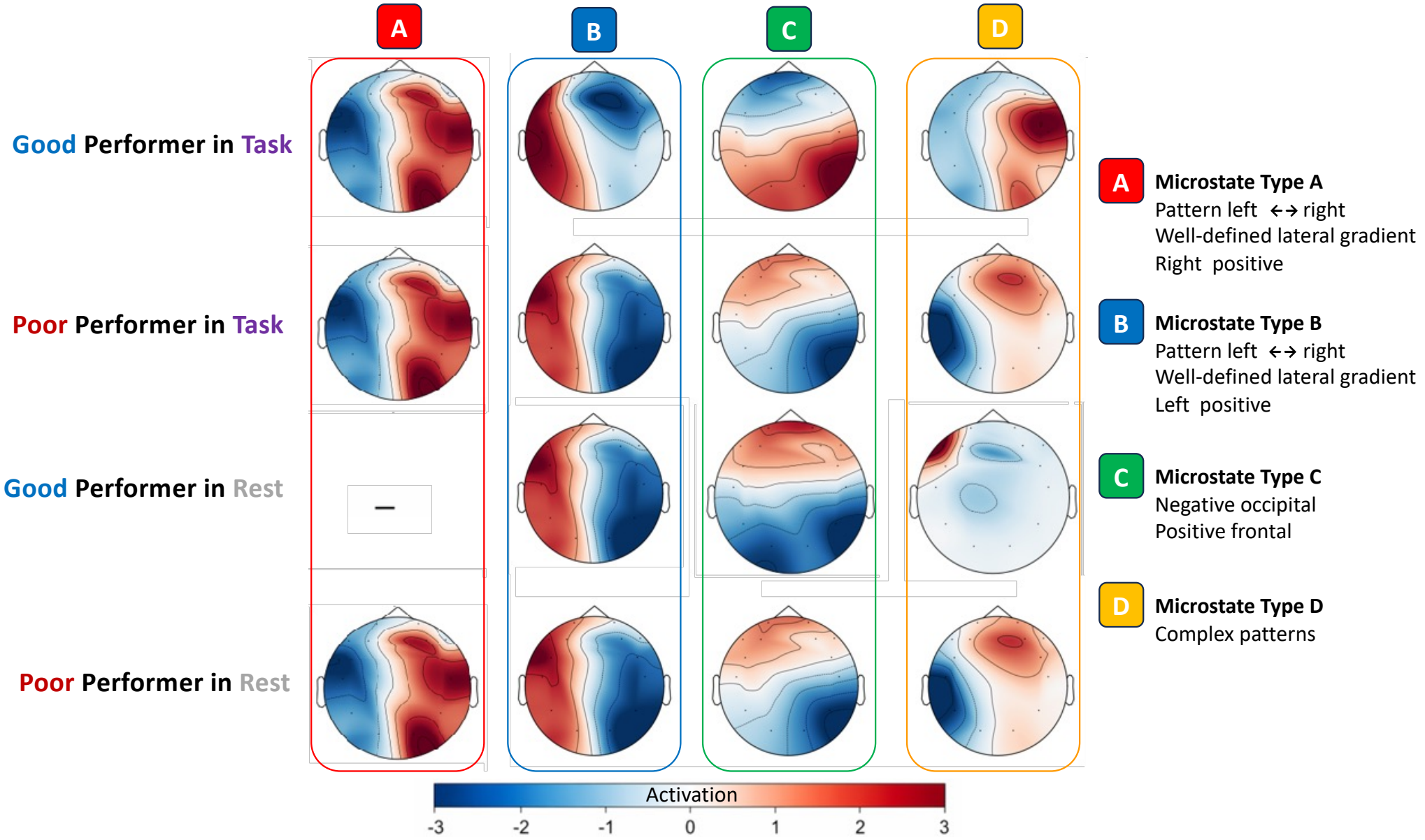





The color maps show brain activation patterns, where red indicates higher activation and blue indicates lower activation in different scalp regions.




Developers who correctly completed the tasks exhibited distinct microstate patterns, particularly in Microstates A, B, and D, whereas poorer performance was more associated with Microstate C.


A
B
C
D
Good Performer in Task
Poor Performer in Task
Good Performer in Rest
Poor Performer in Rest
Activation
-3
-2
-1
0
1
2
3
A
Microstate Type A
Pattern left ↔ right
Well-defined lateral gradient
Right positive
B
Microstate Type B
Pattern left ↔ right
Well-defined lateral gradient
Left positive
C
Microstate Type C
Negative occipital
Positive frontal
D
Microstate Type D
Complex patterns

# Closing Remarks

- Software merge is not only a technical task but fundamentally a human-centered activity.
- Context-aware systems and LLMs can assist developers by anticipating conflicts, explaining changes, and recommending appropriate resolutions.
- Understanding developers' cognitive processes can help the design of more human-centered software engineering tools.

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# Thank you for watching!

# Understanding Software Merge through Empirical Studies and EEG Data

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